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Anti-allergen antibodies

Abstract

The present invention generally relates to antibodies or binding fragments thereof capable of binding an allergen, in particular a food allergen as well as pharmaceutical compositions comprising such antibodies or binding fragments thereof for the treatment of allergy, in particular food allergy. In addition the invention relates to methods for evaluating the capacity of a candidate antibody or binding fragment thereof to inhibit allergen binding/and/or allergen-induced activity in a human and methods of detecting or quantifying whether an allergen is present in a sample.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/625,085, § 371(c) Date: Dec. 20, 2019, which is a national phase entry of International Application No. PCT/EP2018/066430, filed Jun. 20, 2018, which claims the priority benefit of EP Application No. 18160974.4, filed Mar. 9, 2018, and GB Application No. 1710059.5, filed Jun. 23, 2017, each of which is hereby incorporated by reference herein in its entirety.

REFERENCE TO A SEQUENCE LISTING SUBMITTED ELECTRONICALLY VIA EFS-WEB
(1) The content of the electronically submitted sequence listing (Name: 4532_0010003_Seqlisting_ST25.txt; Size: 59,623 bytes; and Date of Creation: Mar. 31, 2022) is herein incorporated by reference in its entirety.
FIELD OF THE INVENTION
(2) The present invention generally relates to antibodies or binding fragments thereof capable of binding an allergen, in particular a food allergen as well as pharmaceutical compositions comprising such antibodies or binding fragments thereof for the treatment of allergy, in particular food allergy. In addition the invention relates to methods for evaluating the capacity of a candidate antibody or binding fragment thereof to inhibit allergen binding/and/or allergen-induced activity in a human and methods of detecting or quantifying whether an allergen is present in a sample.
BACKGROUND OF THE INVENTION
(3) Food allergies are reactions to ingested foods that may lead to clinical manifestations from skin, respiratory and gastrointestinal symptoms up to severe and life-threatening reactions, i.e. systemic anaphylaxis. Allergic reactions to many different types of foods have been described, but some of the most common are reactions to peanuts and shellfish. Such immediate hypersensitivity reactions are caused by immune responses to environmental antigens, also called allergens. Allergen encounter results in the production of immunoglobulin E (IgE) antibodies by antibody-producing cells of the blood, the so-called B cells. IgE antibodies specific for the allergens bind to FcεRI receptors on mast cells and basophils, and upon critical food intake the allergens are binding to pre-bound IgE on the surface of mast cells and basophils. When these cell-associated IgE antibodies are cross-linked by the allergen, mast cells and basophils are activated to rapidly release a variety of

mediators (e.g. histamine, leukotriene, lipid mediators), thereby triggering symptoms of allergy.
(4) Systemic anaphylaxis, as manifested by urticaria, angioedema, bronchospasm, diarrhea, dysrhythmias, and cardiovascular collapse, is responsible for a large number of emergency visits, hospital admissions, and deaths each year. Food allergy is responsible for 20-30% of the emergency visits for anaphylaxis and 100-200 deaths annually in the United States and 50-60% of these are caused by peanut allergy.
(5) Indeed, allergy to peanuts is the most frequent cause of food-related death. Compared to other food allergies, peanut allergy is less likely to be outgrown. Individuals may be so sensitive to peanut allergens that severe systemic reactions can occur in response to minute contaminants of the allergen introduced accidentally during food preparation. Accidental ingestions of peanuts may result in severe and fatal reactions and occur in up to 25-75% of patients over a 5-year period, causing significant anxiety in patients and in families of children with peanut allergy.
(6) Ara h 2 was described as the most important peanut allergen, as it was identified as a predictor of clinical reactivity to peanut. Polysensitization to Ara h 2 and Ara h 1 and/or Ara h 3 appeared to be predictive of more severe reactions (Bublin, M. et al. Cross-reactivity of peanut allergens. Curr. Allergy Asthma Rep. 14, 426 (2014)). In addition, Ara h 6 has recently emerged as a major peanut allergen (Codreanu F, et al. A novel immunoassay using recombinant allergens simplifies peanut allergy diagnosis. Int. Arch. Allergy Immunol. 154, 216-226 (2011)). Ara h 1 (cupin) contributes to 12-16% of the total protein content of a peanut; Ara h 2 (2S albumin) to 5.9-9.3%. Together, Ara h 1, 2 and 3 account for 75% of total protein content. All known peanut allergen classes comprise 85% of the total peanut protein content (Bublin et al.). Mono-sensitization to a single peanut allergen is very rare, but sensitization occurs with a high degree of heterogeneity to a number of peanut allergens (Shreffler, W. G., et al. Microarray immunoassay: Association of clinical history, in vitro IgE function, and heterogeneity of allergenic peanut epitopes. J. Allergy Clin. Immunol. 113, 776-782 (2004)).
(7) Allergen immunotherapy, also known as desensitization, involves exposing the patient to increasing amounts of allergen and is commonly used to treat patients allergic to many seasonal and perennial allergens. In contrast, in peanut allergic patients allergen immunotherapy has not been established, primarily due to the safety risk of treating patients with highly allergenic peanut proteins. In contrast to symptomatic medication such as epinephrine, antihistamines, β-adrenergic agonist and corticosteroid, effective allergen immunotherapy causes an increased immunological tolerance to the allergen or leads to a protective immunity against allergens.

OBJECTIVES AND SUMMARY OF THE INVENTION

(8) In order to meet the above needs, it is one objective of the invention to provide antibodies or binding fragments thereof for the effective treatment of allergies, in particular food allergies, such as peanut allergy.
(9) The antibodies or binding fragments thereof described herein may be capable of reducing, inhibiting or neutralizing allergen-mediated biological activity. In particular, the antibodies may be capable of reducing or inhibiting the binding of an IgE antibody to the food allergen. Thus, the antibodies or binding fragments thereof according to the invention decrease or inhibit the activation of the mast cells or basophils and therefore decrease or prevent the release of mediators (e.g. histamine, lipid mediators, leukotriene). Thereby, the antibodies described herein may inhibit allergy symptoms that would usually occur in the patient after contact with the allergen (e.g. contact with the eyes, nose or mouth or food uptake).
(10) The inventors found that treatment of patients suffering from peanut allergy with human-derived antibodies could be highly effective. Since human-derived antibodies underwent the natural “evolution” in the human body, these antibodies are highly effective for the treatment of a patient.
(11) The food allergen may be a peanut allergen. In an specific embodiment the peanut allergen is selected from the group consisting of Ara h 1, Ara h 2, Ara h 3, Ara h 4, Ara h 5, Ara h 6/7, Ara h 8, Ara h 9 and Ara h 10/11. Preferably, the allergen is selected from the group consisting of Ara h 1,

Ara h 2, Ara h 3 and Ara h 6, or a combination thereof. Typically, the peanut allergen is of peanut origin, recombinantly expressed or is a synthetic peanut peptide.
(12) Typically, the antibody may be a monoclonal antibody. In some embodiments, the antibody may be a recombinant antibody.
(13) The antibody may be an IgG or IgA antibody. IgG and IgA compete with IgE for binding sites on the allergen and thereby prevent recognition of allergens by IgE bound to Fcε receptors on the surface of mast cells and basophils. This may include direct competition by binding to the same epitope or competition through steric hindrance. Furthermore, IgG antibodies bound to the allergen can lead to cross-linking of Fcε and inhibitory FcγRIIB receptors, resulting in the decrease of effector cell activity. Thereby the IgG and IgA antibodies or binding fragments thereof according to the invention can be used for the effective prevention or treatment of allergies.
(14) In some embodiments, the variable regions, portions thereof or the CDRs are human-derived. For example, the variable regions, portions thereof or the CDRs are derived from an IgE antibody and grafted in a scaffold of an IgG or IgA antibody. In one embodiment, the peptide sequence of the antibody or binding fragment thereof is identical or at least 60% identical to the sequence of the antibody extracted from the human. The human from which the antibody is extracted may be a human suffering from peanut allergy, a peanut-sensitized human without clinical relevant allergy, a human suffering from peanut allergy that underwent immunotherapy, a human that has outgrown peanut allergy and a human of unknown clinical history for peanut allergy.
(15) In some embodiments, the antibody or binding fragment thereof has an EC50 of at most 270 ng/ml, preferably at most 70 ng/ml, at most 40 ng/ml, at most 25 ng/ml, at most 15 ng/ml, at most 4.9 ng/ml, at most 1.3 ng/ml for at least one of the peanuts allergens selected from the group consisting of Ara h 2, Ara h 1, Ara h 3 and Ara h 6. Preferably, the antibody has an EC50 of at most 10 ng/ml, preferably at most 7 ng/ml, more preferably at most 4.8 ng/ml, most preferably at most 2.8 ng/ml for peanut extract.
(16) The antibodies described herein may be capable of reducing, inhibiting or neutralizing allergen-mediated biological activity. In particular, the antibodies may be capable of reducing or inhibiting the binding of an IgE antibody to the food allergen.
(17) In one embodiment, the antibody or binding fragment thereof comprises a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 1 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 2 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 3 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 4 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 5 or sequences at least 65% identical thereto, and a CDR3 set forth in SEQ ID No: 6 or sequences at least 65% identical thereto.
(18) In another embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 15 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 16 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 17 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 18 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 19 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 20 or sequences at least 65% identical thereto.
(19) A further embodiment refers to an antibody or binding fragment comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 29 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 30 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 31 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 32 or sequences at least 65% identical thereto, a CDR2

set forth in SEQ ID No: 33 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 34 or sequences at least 65% identical thereto.

(20) A further embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 43 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 44 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 45 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 46 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 47 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 48 or sequences at least 65% identical thereto.

(21) A further embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 57 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 58 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 59 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 60 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 61 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 62 or sequences at least 65% identical thereto.

(22) A further embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 71 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 72 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 72 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 74 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 75 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 76 or sequences at least 65% identical thereto.

(23) The antibodies are described in more detail below.

(24) Preferably, the antibody is selected from the group consisting of a full length antibody, multispecific antibody, a single chain Fv fragment (scFv), a F(ab′) fragment, a F(ab′)2 fragment, a F(ab)c fragment, a single domain antibody fragment (sdAB) or a multispecific antibody fragment.

(25) Another aspect of the invention refers to an antibody or binding fragment thereof which competes with an antibody as described herein for specific binding to the allergen.

(26) A further aspect refers to a polynucleotide encoding the antibody or binding fragment thereof as described herein. Another embodiment refers to a vector comprising a polynucleotide encoding the antibody or binding fragment thereof as described herein. Another embodiment refers to a cell comprising a polynucleotide encoding the antibody or binding fragment thereof as described herein or a vector comprising a polynucleotide encoding the antibody or binding fragment thereof as described.

(27) A further aspect refers to a method for preparing an anti-allergen antibody or allergen-binding fragment thereof, consisting of culturing the cell comprising a polynucleotide encoding the antibody or binding fragment thereof as described herein and isolating the antibody or allergen binding fragment thereof from the cell or culture medium of the cell.

(28) Moreover, the invention relates to an antibody composition comprising at least two antibodies, wherein at least one of the antibodies is selected from the antibodies as defined herein. In one embodiment the antibody composition comprises at least two of the antibodies selected from the antibodies as defined herein. In one embodiment the antibody composition comprises at least three of the antibodies selected from the antibodies as defined herein.

(29) Another aspect of the invention refers to a pharmaceutical composition comprising at least one of the compounds selected from the group consisting of antibody or binding fragment as described herein, the antibody composition as described herein, the polynucleotide as described herein, the

vector as described herein and the cell as described herein.
(30) In one embodiment, the pharmaceutical comprises the antibody or binding fragment of any as described herein or the antibody composition as described herein.
(31) Another embodiment the pharmaceutical composition further comprises an additional agent useful for treating peanut allergy. The additional agent useful for treating peanut allergy may be a β-adrenergic agonist (e.g. epinephrine), antihistamine, corticosteroid, anti-IgE antibody, anti-IgE antibody binding fragment, peptide vaccine and further antibodies capable of binding to a peanut allergen. Preferably, the pharmaceutical composition comprises epinephrine.
(32) Typically, the pharmaceutical composition may comprise a pharmaceutical acceptable carrier.
(33) Preferably, the pharmaceutical composition is intended for subcutaneous, intravenous, intramuscular, intraperitoneal, intranasal and/or inhalative administration. The antibody or binding fragment thereof, in particular the pharmaceutical composition comprising the antibody or binding fragment, may be administered once or more times.
(34) Other aspects of the invention refer to the antibodies or binding fragments, antibody composition, polynucleotide, vector, cell or pharmaceutical composition as described herein for use in the treatment of allergy. The allergy may be a food allergy, such as peanut allergy.
(35) In particular, the invention refers to the antibodies or binding fragments as described herein, the antibody composition as described herein and the pharmaceutical composition as described herein for use in the treatment of food allergy, such as peanut allergy. The treatment may be prophylactic or therapeutic.
(36) Another aspect refers to a method of evaluating the capacity of a candidate antibody or binding fragment thereof to inhibit allergen binding/and/or allergen-induced activity in a human,
(37) wherein the method comprises
(38) (i) incubating the candidate antibody or binding fragment thereof with a composition comprising IgEs derived from the human and a food allergen;
(39) (ii) evaluating whether the candidate antibody inhibits allergen binding/and/or allergen-induced activity in the composition comprising IgEs derived from the human.
(40) Thus, the method of the invention allows to tailor the therapy to the patient based on the predicted response to the antibodies or fragments thereof according to the invention. This approach is also referred as “personalized medicine” approach thereby allows to identify the most effective antibody or antibody combination.
(41) In one embodiment the method further comprises the step of
(42) (iii) determining whether the administration of the candidate antibody is a suitable treatment for a patient suffering from food allergy based on the result of step (ii).
(43) In other words, the present application refers to methods for determining whether a patient suffering from food allergy is responsive to a candidate antibody, comprising the steps of
(44) (i) incubating the candidate antibody or binding fragment thereof with a composition comprising IgEs derived from the human and a food allergen;
(45) (ii) evaluating whether the candidate antibody inhibits allergen binding and/or allergen-induced activity in the composition comprising IgEs derived from the human;
(46) (iii) determining whether the administration of the candidate antibody is a suitable treatment for the patient suffering from food allergy based on the result of step (ii).
(47) Preferably the candidate antibody is an antibody of the invention as described herein.
(48) Preferably, in step (i) the composition comprising IgEs from the allergic patient comprises basophils. Typically, the basophils are derived from the patient. Alternatively, the basophils are donor-derived IgE-stripped basophils.
(49) In one embodiment, basophils are identified based on the low side-scattered light (SSC) and the expression of CCR3. In addition, activated basophils are identified by measuring surface expression of CD63 and/or surface expression of CD203c. Typically, in activated basophils the surface expression of CD63 and/or CD203c is high.

(50) In another embodiment, the secretion of a mediator from leukocytes, in particular basophils, is measured. In a preferred embodiment the mediator is leukotriene, such as sulfidoleukotriene.
(51) The measurement of secreted mediators, in particular leukotriene is advantageous, since the measurement of the mediators, preferably leukotriene, released from the leukocytes, in particular basophils, can be detected by ELISA and therefore can be easily carried out in high-throughput format. Leukotriene is released from the basophils in a dose dependent manner. Thus, it is particularly suitable for highly sensitive assays.
(52) The composition comprising IgEs derived from the human may be plasma, sera, blood, saliva, peripheral blood mononuclear cell (PBMC), leukocytes, basophils or IgEs stripped from basophils.
(53) When measuring secreted mediators, preferably leukotriene, leukocytes may be used as composition comprising IgEs. This makes the assay straight forward, since it is not necessary to distinguish or isolate the basophils from the other cell fractions (such as neutrophils, eosinophils, lymphocytes and monocytes) of the leukocyte pool for measuring secreted mediators.
(54) Another aspect of the invention refers to a method of detecting or quantifying whether an allergen is present in a sample comprising the following steps:
(55) i) incubation of the sample with an antibody of the invention as described herein or with an antibody composition of the invention as described herein,
(56) ii) detecting the antibody which is bound to allergen in the sample.
(57) The antibody may be detectably labeled. Alternatively, the antibody is used with a second antibody that is detectably labeled. Preferably, the antibody is unlabeled and used in combination with a second antibody that is detectably labeled. The detectable label may be selected from the group consisting of an enzyme, a radioisotope, a fluorophore, a peptide and a heavy metal.

Description

FIGURE LEGENDS
(1) FIG. **1**. histogram distribution of the 84 allergic patients enrolled in the clinical trial described in example 1, total IgG, IgG4, IgE antibody reactivities against major peanut allergens (Ara h 1, 2 and 3). Seroreactivities are expressed as titer, which is defined as OD450 value over background.
(2) FIG. **2**. histogram distribution of the 84 allergic patients enrolled in the clinical trial described in example 1, total IgG, IgG4, IgE antibody reactivities against major peanut allergen Ara h 6 and peanut extract. Seroreactivities are expressed as titer, which is defined as OD450 value over background.
(3) FIG. **3**. EC50 ELISA determination of binding of exemplary anti-Ara h 2 antibody 32B10 to naturally extracted peanut allergens (nAra h 1, nAra h 2, nAra h 3, nAra h 6, Indoor Biotechnologies, Cardiff, UK), recombinantly expressed Ara h 2 (rAra h 2, Indoor Biotechnologies, Cardiff, UK) and natural peanut extract. Bovine serum albumin (BSA) serves as control. Similar binding was detected for the antibodies 37D5, 12G10, 4B2, 2F8 and 7G6.
(4) FIG. **4**. Human-derived anti-peanut allergen monoclonal antibodies inhibit allergen-mediated activation of basophils derived from allergic patients. Basophils were either left untreated or stimulated with peanut extract in the presence of human-derived antibodies or an isotype control. A. Bars indicate activation of basophils expressed as % CD63+ cells (mean±s.d, n=3). Pre-incubation of peanut extract with exemplary antibodies (32B10, 37D5, 12G10, 4B2) or a combination of them (MY006 cocktail) decreased basophil degranulation, whereas isotype control did not have any effect. White bars: no addition of antibody.
(5) FIG. **5**. Human-derived anti-peanut allergen monoclonal antibodies inhibit allergen-mediated release of leukotriene (sLT) from leukocytes derived from allergic patients. Leukocytes were stimulated with increasing concentrations of peanut extract (PE; 0-30 ng/ml) in the presence of a cocktail of human-derived antibodies (MY006) or an isotype control. Pre-incubation of peanut

extract with MY006 cocktail inhibited leukotriene release (curve shift to the right), whereas isotype control did not have any effect.

(6) FIG. **6**. Effect of different anti-peanut allergen antibody combinations can be evaluated by leukotriene (sLT) release assay. Leukocytes from healthy blood donors were isolated and surface IgE was removed by incubation with lactic acid. Leukocytes were re-sensitized with IgE by incubation with plasma from allergic patients. Re-sensitized leukocytes were stimulated with peanut extract (0-30 ng/ml) in the presence of different human-derived monoclonal anti-peanut allergen antibodies or an isotype control. Pre-incubation of peanut extract (PE) with human-derived anti-peanut allergen antibodies inhibited leukotriene release to different extents, whereas isotype control did not have any effect.

DETAILED DESCRIPTION OF THE INVENTION

(7) Before the invention is described in detail with respect to some of its preferred embodiments, the following general definitions are provided.

(8) The present invention as illustratively described in the following may suitably be practiced in the absence of any element or elements, limitation or limitations, not specifically disclosed herein.

(9) The present invention will be described with respect to particular embodiments and with reference to certain figures but the invention is not limited thereto but only by the claims.

(10) Where the term “comprising” is used in the present description and claims, it does not exclude other elements. For the purposes of the present invention, the term “consisting of” is considered to be a preferred embodiment of the term “comprising of”. If hereinafter a group is defined to comprise at least a certain number of embodiments, this is also to be understood to disclose a group which preferably consists only of these embodiments.

(11) For the purposes of the present invention, the term “obtained” is considered to be a preferred embodiment of the term “obtainable”. If hereinafter e.g. an antibody is defined to be obtainable from a specific source, this is also to be understood to disclose an antibody which is obtained from this source.

(12) Where an indefinite or definite article is used when referring to a singular noun, e.g. “a”, “an” or “the”, this includes a plural of that noun unless something else is specifically stated. The terms “about” or “approximately” in the context of the present invention denote an interval of accuracy that the person skilled in the art will understand to still ensure the technical effect of the feature in question. The term typically indicates deviation from the indicated numerical value of ±10%, and preferably of ±5%.

(13) Technical terms are used by their common sense. If a specific meaning is conveyed to certain terms, definitions of terms will be given in the following in the context of which the terms are used.

(14) A first aspect relates to an antibody or binding fragment thereof capable of binding to a food allergen.

(15) The term “capable of binding to a food allergen” as used herein means that the antibody is specific for the allergen, i.e. it recognizes the allergen. Preferably, the antibody has a higher affinity for the specific allergen, i.e. the specific food allergen, such as the peanut allergen Ara h 1, Ara h 2, Ara h 3, Ara h 6 than for other allergens.

(16) The affinity or avidity of an antibody for an antigen can be determined experimentally using any suitable method; see, for example, Berzofsky et al., “Antibody-Antigen Interactions” In Fundamental Immunology, Paul, W. E., Ed., Raven Press New York, N Y (1984), Kuby, Janis Immunology, W. H. Freeman and Company New York, N Y (1992), and methods described herein. The measured affinity of a particular antibody-antigen interaction can vary if measured under different conditions, e.g., salt concentration, pH. Thus, measurements of affinity and other antigen-binding parameters, e.g., K.sub.D, IC50, EC50 are preferably made with standardized solutions of antibody and antigen, and a standardized buffer.

(17) Preferably the antibody or binding fragment thereof is human-derived.

(18) The CDRs, the variable and/or the constant regions may stem from the same human-derived

antibody. Alternatively, the variable regions, portions thereof or the CDRs are human-derived and are grafted in an antibody framework. This antibody framework may be of human origin. Typically, the human-derived portions of the variable regions that are grafted into the antibody framework comprise the CDRs.
(19) The antibody or binding fragment thereof may comprise a CL and/or CH constant region comprising an amino acid sequence selected from the CL amino acid sequences SEQ ID NOS: 9, 23, 37, 51 and 65 or an amino acid sequence with at least 60% identity and an amino acid sequence selected from the CH amino acid sequences SEQ ID NOS: 10, 24, 38, 52 and 66 or an amino acid sequence with at least 60% identity.
(20) The term “allergen” in the context of the invention is a compound that is recognized by the immune system as foreign so that an immunoreaction to the allergen is evoked. In other words it refers to molecules that are not present in the human body and may produce an abnormal immune response by the activation of mast cells and basophils triggering symptoms of allergy. The skilled person understands that in the context of the invention “autoantigens”, i.e. molecules that are produced by the human body are not considered as allergen.
(21) The antibody may be capable of binding a food allergen. Food allergen may be any food that that causes allergy or ingredient thereof, in particular protein. The food allergen may be for example milk, eggs, crustacean, shellfish, tree nuts, peanuts, wheat or soybean or ingredients thereof. In one embodiment the food allergen is peanut or tree nut. In a specific embodiment the food allergen is a legume, such as soybean or peanut. Patients with peanut allergy may have a greater chance of being allergic to other legumes such as soy.
(22) In one embodiment, the allergen is a peanut allergen. The peanut allergen may be untreated or treated peanut, such as roasted peanut. The peanut allergen may be a peanut protein, such as Ara h 1, Ara h 2, Ara h 3, Ara h 4, Ara h 5, Ara h 6/7, Ara h 8, Ara h 9 and Ara h 10/11. In a specific embodiment the peanut protein may be in particular selected from Ara h 1, Ara h 2, Ara h 3 and Ara h 6, or a combination thereof. The peanut allergen may be of peanut origin, recombinantly expressed or is a synthetic peanut peptide.
(23) Typically, the antibody is a monoclonal antibody. In some embodiments, the antibody is a recombinant antibody.
(24) Antibody or binding fragment thereof according to any one of the preceding claims, wherein the antibody is an IgG, such as an IgG1, IgG2, IgG3 or IgG4 antibody, or IgA antibody.
(25) In one embodiment the IgG may be an IgG1, IgG2 or IgG4 antibody.
(26) The term “human-derived” in the context of the present invention means that at least the CDRs are derived from a human antibody. The “human-derived” antibody may contain further elements that are derived from the human antibody, such as parts or the complete framework of the heavy and/or light chain variable regions and/or the parts or the complete of the heavy and/or light chain constant region. For example, the variable regions, portions thereof or the CDRs may be derived from an IgE antibody and grafted in a scaffold of an IgG or IgA antibody. The peptide sequence of the human-derived antibody may be at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% or at least 97% identical to the sequence of the antibody extracted from the human. Preferably, the peptide sequence of the human-derived antibody may be at least 90%, at least 95%, at least 97%, at least 98% or at least 99% identical to the sequence of the antibody extracted from the human.
(27) The human from which the antibody is extracted may be selected from the group of a human suffering from peanut allergy, a peanut-sensitized human without clinical relevant allergy, a human suffering from peanut allergy that underwent immunotherapy, a human that has outgrown peanut allergy and a human of unknown clinical history for peanut allergy.
(28) The antibodies of the invention may be used for the prevention or treatment of allergy.
(29) Allergy symptoms may be skin rash, itching skin, itching or tingling sensation in or around the mouth or throat, headache, sneezing, swelling, nausea, diarrhea or anaphylaxis.

(30) In some embodiments, the antibody or binding fragment thereof has an EC50 of at most 270 ng/ml, preferably at most 70 ng/ml, at most 40 ng/ml, at most 25 ng/ml, at most 15 ng/ml, at most 4.9 ng/ml, at most 1.3 ng/ml for at least one of the peanut allergens selected from the group consisting of Ara h 2, Ara h 1, Ara h 3 and Ara h 6. Preferably, the antibody has an EC50 of at most 10 ng/ml, preferably at most 7 ng/ml, more preferably at most 4.8 ng/ml, most preferably 2.8 ng/ml for peanut extract. The EC50 may be measured by an ELISA assay as described herein.
(31) In one embodiment, the antibody or binding fragment thereof comprises a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 1 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 2 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 3 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 4 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 5 or sequences at least 65% identical thereto, and a CDR3 set forth in SEQ ID No: 6 or sequences at least 65% identical thereto.
(32) Another embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 15 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 16 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 17 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 18 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 19 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 20 or sequences at least 65% identical thereto.
(33) A further embodiment refers to an antibody or binding fragment comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 29 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 30 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 31 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 32 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 33 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 34 or sequences at least 65% identical thereto.
(34) A further embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 43 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 44 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 45 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 46 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 47 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 48 or sequences at least 65% identical thereto.
(35) A further embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 43 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 44 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 45 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 46 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 47 or sequences at least 75% identical thereto, a CDR3 set forth in SEQ ID No: 48 or sequences at least 65% identical thereto.
(36) A further embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 57 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 58 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID

No: 59 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 60 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 61 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 62 or sequences at least 65% identical thereto.

(37) A further embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 71 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 72 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 73 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 74 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 75 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 76 or sequences at least 65% identical thereto.

(38) One embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 1 or sequences at least 91% identical thereto, a CDR2 set forth in SEQ ID No: 2 or sequences at least 71% identical thereto, a CDR3 set forth in SEQ ID No: 3; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 4 or sequences at least 80% identical thereto, a CDR2 set forth in SEQ ID No: 5 or sequences at least 82% identical thereto, a CDR3 set forth in SEQ ID No: 6 or a sequence at least 94% identical thereto.

(39) A further embodiment refers to an antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 71 or sequences at least 91% identical thereto, a CDR2 set forth in SEQ ID No: 72 or sequences at least 71% identical thereto, a CDR3 set forth in SEQ ID No: 73; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 74 or sequences at least 80% identical thereto, a CDR2 set forth in SEQ ID No: 75 or sequences at least 82% identical thereto, a CDR3 set forth in SEQ ID No: 76 or sequences at least 94% identical thereto.

(40) A further embodiment refers to an antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a sequence set forth in SEQ ID No: 7 or sequences at least 70% identical thereto, and/or wherein the heavy chain variable region comprises a sequence set forth in SEQ ID No: 8 or sequences at least 70% identical thereto.

(41) A further embodiment refers to an antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a sequence set forth in SEQ ID No: 21 or sequences at least 70% identical thereto, and/or wherein the heavy chain variable region comprises a sequence set forth in SEQ ID No: 22 or sequences at least 70% identical thereto.

(42) A further embodiment refers to an antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a sequence set forth in SEQ ID No: 35 or sequences at least 70% identical thereto, and/or wherein the heavy chain variable region comprises a sequence set forth in SEQ ID No: 36 or sequences at least 70% identical thereto.

(43) A further embodiment refers to an antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a sequence set forth in SEQ ID No: 49 or sequences at least 70% identical thereto, and/or wherein the heavy chain variable region comprises a sequence set forth in SEQ ID No: 50 or sequences at least 70% identical thereto.

(44) A further embodiment refers to an antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region

comprises a sequence set forth in SEQ ID No: 63 or sequences at least 70% identical thereto, and/or wherein the heavy chain variable region comprises a sequence set forth in SEQ ID No: 64 or sequences at least 70% identical thereto.

(45) A further embodiment refers to an antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a sequence set forth in SEQ ID No: 77 or sequences at least 70% identical thereto, and/or wherein the heavy chain variable region comprises a sequence set forth in SEQ ID No: 78 or sequences at least 70% identical thereto.

(46) A further embodiment refers to an antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a sequence set forth in SEQ ID No: 7 or sequences at least 91% identical thereto, and/or wherein the heavy chain variable region comprises a sequence set forth in SEQ ID No: 8 or sequences at least 88% identical thereto.

(47) A further embodiment refers to an antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a sequence set forth in SEQ ID No: 77 or sequences at least 91% identical thereto, and/or wherein the heavy chain variable region comprises a sequence set forth in SEQ ID No: 78 or sequences at least 88% identical thereto.

(48) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region at least one CDR of the light chain variable region comprising the amino acid sequence SEQ ID NO: 7 and/or the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 8.

(49) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region the CDR1, CDR2 and CDR3 of the light chain variable region comprising the amino acid sequence SEQ ID NO: 7 and the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 8.

(50) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region at least one CDR of the light chain variable region comprising the amino acid sequence SEQ ID NO: 21 and/or the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 22.

(51) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region the CDR1, CDR2 and CDR3 of the light chain variable region comprising the amino acid sequence SEQ ID NO: 21 and the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 22.

(52) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region at least one CDR of the light chain variable region comprising the amino acid sequence SEQ ID NO: 35 and/or the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 36.

(53) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region the CDR1, CDR2 and CDR3 of the light chain variable region comprising the amino acid sequence SEQ ID NO: 35 and the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 36.

(54) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region at least one CDR of the light chain variable region comprising the amino acid sequence SEQ ID NO: 49 and/or the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 50.

(55) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region the CDR1, CDR2 and CDR3 of the light chain variable region comprising the amino acid sequence SEQ ID NO: 49 and the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 50.

(56) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region at least one CDR of the light chain variable region comprising the amino acid sequence SEQ ID NO: 63 and/or the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 64.
(57) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region the CDR1, CDR2 and CDR3 of the light chain variable region comprising the amino acid sequence SEQ ID NO: 63 and the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 64.
(58) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region at least one CDR of the light chain variable region comprising the amino acid sequence SEQ ID NO: 77 and/or the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 78.
(59) A further embodiment refers to an antibody or binding fragment thereof, comprising in its variable region the CDR1, CDR2 and CDR3 of the light chain variable region comprising the amino acid sequence SEQ ID NO: 77 and the heavy chain variable region comprising the amino acid sequence SEQ ID NO: 78.
(60) The person skilled in the art knows that each variable domain (the heavy chain V.sub.H and light chain V.sub.L) of an antibody comprises three hypervariable regions, sometimes called complementarity determining regions or "CDRs" flanked by four relatively conserved framework regions or "FRs" and refer to the amino acid residues of an antibody which are responsible for antigen-binding. Exemplary conventions that can be used to identify the boundaries of CDRs include, e.g., the Kabat definition, the Chothia definition, and the AbM definition. In general terms, the Kabat definition is based on sequence variability, the Chothia definition is based on the location of the structural loop regions, and the AbM definition is a compromise between the Kabat and Chothia approaches. See, e.g., Kabat, "Sequences of Proteins of Immunological Interest," National Institutes of Health, Bethesda, Md. (1991); Al-Lazikani et al., (1997), J. Mol. Biol. 273:927-948; and Martin et al., (1989), Proc. Natl. Acad. Sci. USA 86:9268-9272. For example, the hypervariable regions or CDRs of the human IgG subclass of antibody comprise amino acid residues from residues 24-34 (L1), 50-56 (L2) and 89-97 (L3) in the light chain variable domain and 31-35 (H1), 50-65 (H2) and 95-102 (H3) in the heavy chain variable domain as described by Kabat et al., and/or those residues from a hypervariable loop, i.e. residues 26-32 (L1), 50-52 (L2) and 91-96 (L3) in the light chain variable domain and 26-32 (H1), 53-55 (H2) and 96-101 (H3) in the heavy chain variable domain as described by Chothia et al., J. Mol. Biol. 196 (1987), 901-917. Framework or FR residues are those variable domain residues other than and bracketing the hypervariable regions. Accordingly, the term "CDR" refers to the complementarity determining region or hypervariable region amino acid residues of an antibody that participate in or are responsible for antigen-binding. The CDRs as described herein are defined according to Kabat et al. (1991) as described in *Sequences of Proteins of Immunological Interest.*
(61) Binding fragments may thus include portions of an intact full length antibody, such as an antigen binding or variable region of the complete antibody. Examples of antibody fragments include F(ab′), F(ab′)2, F(ab)c, and Fv fragments; diabodies; linear antibodies; single-Fv fragments (e.g., scFv); multispecific antibody fragments such as bispecific, trispecific, and multispecific antibodies (e.g., diabodies, triabodies, tetrabodies); minibodies; chelating recombinant antibodies; tribodies or bibodies; intrabodies; nanobodies; small modular immunopharmaceuticals (SMIP), binding-domain immunoglobulin fusion proteins; camelized antibodies; VHH containing antibodies; chimeric antigen receptor (CAR); and any other polypeptides formed from antibody fragments. The skilled person is aware that the antigen-binding function of an antibody can be performed by fragments of a full-length antibody. The binding fragment may have a length of at least 5, at least 8, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15 or more amino acids. The fragment may have the antigen-binding function of the antibody.

(62) The skilled person is aware that also binding molecules capable of binding to a food allergen, in particular a peanut allergen and epitopes thereof, as defined herein, are encompassed by the invention. In particular encompassed are binding molecules capable of binding to a food allergen as defined herein and capable of preventing recognition of food allergens by IgE bound to FCC receptors. Hence the invention also includes all aspects relating to these binding molecules, such as pharmaceutical compositions, kits, therapeutic and diagnostic uses as well as methods relating to these binding molecules. Thus, the term "binding molecule" includes not only antibodies and binding fragments thereof but also includes non-antibody protein scaffold drugs, such as DARPins (for example reviewed in "Challenges and opportunities for non-antibody scaffold drugs", Vazquez-Lombardi R. et al Drug Discovery Today 20 (10): 1271-1283) and other molecules that are not related to peptide structures.
(63) The term "antibody" and "immunoglobulin" are used interchangeably herein. Preferably, the antibody is a full length antibody, multispecific antibody, a single chain Fv fragment (scFv), a F(ab′) fragment, a F(ab′)2 fragment, a F(ab)c fragment, a single domain antibody fragment (sdAB) or a multispecific antibody fragment.
(64) A Fab fragment consists of the V.sub.L, V.sub.H, C.sub.L and C.sub.H1 domains. An F(ab′)2 fragment comprises two Fab fragments linked by a disulfide bridge at the hinge region. An Fd is the V.sub.H and C.sub.H1 domains of a single arm of an antibody. An Fv fragment is the V.sub.L and V.sub.H domains of a single arm of an antibody. An F(ab′)c fragment comprises two F(ab′) fragments plus part of the Fc domain. It is generated e.g. by plasmin digestion.
(65) Binding fragments also encompass monovalent or multivalent, or monomeric or multimeric (e.g. tetrameric), CDR-derived binding domains.
(66) A bispecific antibody comprises two different binding specificities and thus binds to two different antigens or two different epitopes of the same antigen. In one embodiment, the bispecific antibody comprises a first antigen recognition domain that binds to a first antigen and a second antigen recognition domain that binds to a second antigen. In another embodiment the bispecific antibody comprises a first antigen recognition domain that binds to a first epitope of the antigen and a second antigen recognition domain that binds to a second epitope of the antigen.
(67) The determination of percent identity between multiple sequences is preferably accomplished using the AlignX application of the Vector NTI Advance™ 10 program (Invitrogen Corporation, Carlsbad CA, USA). This program uses a modified Clustal W algorithm (Thompson et al., 1994. Nucl Acids Res. 22:pp. 4673-4680; Invitrogen Corporation; Vector NTI Advance™ 10 DNA and protein sequence analysis software. User's Manual, 2004, pp.389-662). The determination of percent identity is performed with the standard parameters of the AlignX application.
(68) In these embodiments the sequence identity may be at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95% or at least 97%. Preferably, in all these embodiments the sequence identity is at least about 85%, more preferably at least about 90%, even more preferably at least about 95% and most preferably at least about 98% or about 99%. Sequence identity may be determined over the whole length of the respective sequences.
(69) Another aspect of the invention refers to an antibody or binding fragment thereof which competes with an antibody as described herein for specific binding to the allergen. The skilled person is aware of methods for testing the capability of the antibody or binding fragment thereof to compete for specific binding to the allergen, such as the competition ELISA assay as described elsewhere herein.
(70) Some embodiments refer to an antibody which is capable of binding to at least one of the Ara h 2 epitopes selected from the group consisting of the amino acid sequences set forth in SEQ ID NOs: 84, 88, 89, 90 and 91 or fragments thereof.
(71) A particular embodiment refers to an antibody which is capable of binding to the Ara h 2 epitope defined by the amino acid sequence set forth in SEQ ID NO: 84 or fragments thereof.
(72) Another embodiment refers to an antibody which is capable of binding to at least one of the

Ara h 2 epitopes selected from the group consisting of the amino acid sequences set forth in SEQ ID NOs: 88, 89, 90 and 91 or fragments thereof.
(73) The fragment of the epitope may comprise at least 5, at least 8, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 17 or more amino acids.
(74) Another embodiment refers to an antibody which is capable of binding all of the Ara h 2 epitopes selected from the group consisting of the amino acid sequences set forth in SEQ ID NOs: 88, 89, 90 and 91.
(75) Some embodiments refer to an antibody which is capable of binding to at least one of the Ara h 2 epitopes selected from the group consisting of the amino acid sequences set forth in SEQ ID NOs: 84, 88, 89, 90 and 91.
(76) A particular embodiment refers to an antibody which is capable of binding to the Ara h 2 epitope defined by the amino acid sequence set forth in SEQ ID NO: 84.
(77) Another embodiment refers to an antibody which is capable of binding to at least one of the Ara h 2 epitopes selected from the group consisting of the amino acid sequences set forth in SEQ ID NOs: 88, 89, 90 and 91.
(78) Another embodiment refers to an antibody which is capable of binding to all of the Ara h 2 epitopes defined by the amino acid sequences set forth in SEQ ID NOs: 88, 89, 90 and 91.
(79) The antibody or its encoding cDNAs may be further modified. Thus, in a further embodiment the method of the present invention comprises any one of the step(s) of producing a chimeric antibody, single-chain antibody, Fab-fragment, bi-specific antibody, fusion antibody, labeled antibody or an analog of any one of those. Corresponding methods are known to the person skilled in the art and are described, e.g., in Harlow and Lane “Antibodies, A Laboratory Manual”, CSH Press, Cold Spring Harbor, 1988. When derivatives of said antibodies are obtained by the phage display technique, surface plasmon resonance as employed in the BIAcore system can be used to increase the efficiency of phage antibodies which bind to the same epitope as that of any one of the antibodies described herein (Schier, Human Antibodies Hybridomas 7 (1996), 97-105; Malmborg, J. Immunol. Methods 183 (1995), 7-13). The production of chimeric antibodies is described, for example, in international application WO89/09622. Methods for the production of humanized antibodies are described in, e.g., European application EP-Al 0 239 400 and international application WO90/07861. As discussed above, the antibody of the invention may exist in a variety of forms besides complete antibodies; including, for example, Fv, Fab and F(ab)2, as well as in single chains; see e.g. international application WO88/09344.
(80) The antibodies of the present invention or their corresponding immunoglobulin chain(s) can be further modified using conventional techniques known in the art, for example, by using amino acid deletion(s), insertion(s), substitution(s), addition(s), and/or recombination(s) and/or any other modification(s) known in the art either alone or in combination. Methods for introducing such modifications in the DNA sequence underlying the amino acid sequence of an immunoglobulin chain are well known to the person skilled in the art; see, e.g., Sambrook, Molecular Cloning A Laboratory Manual, Cold Spring Harbor Laboratory (1989) N.Y. and Ausubel, Current Protocols in Molecular Biology, Green Publishing Associates and Wiley Interscience, N.Y. (1994). Modifications of the antibody of the invention include chemical and/or enzymatic derivatizations at one or more constituent amino acids, including side chain modifications, backbone modifications, and N- and C-terminal modifications including acetylation, hydroxylation, methylation, amidation, and the attachment of carbohydrate or lipid moieties, cofactors, and the like. Likewise, the present invention encompasses the production of chimeric proteins which comprise the described antibody or some fragment thereof at the amino terminus fused to heterologous molecule such as an immunostimulatory ligand at the carboxyl terminus; see, e.g., international application WO00/30680 for corresponding technical details.
(81) A further aspect refers to a polynucleotide encoding the antibody or binding fragment thereof as described herein. Another embodiment refers to a vector comprising a polynucleotide encoding

the antibody or binding fragment thereof as described herein. Another embodiment refers to a cell comprising a polynucleotide encoding the antibody or binding fragment thereof as described herein or a vector comprising a polynucleotide encoding the antibody or binding fragment thereof as described.

(82) In this respect, the person skilled in the art will readily appreciate that the polynucleotides encoding at least the variable domain of the light and/or heavy chain may encode the variable domains of both immunoglobulin chains or only one. Likewise, said polynucleotides may be under the control of the same promoter or may be separately controlled for expression. Possible regulatory elements permitting expression in prokaryotic host cells comprise, e.g., the P.sub.L, lac, trp or tac promoter in *E. coli*, and examples for regulatory elements permitting expression in eukaryotic host cells are the AOX1 or GAL1 promoter in yeast or the CMV-, SV40-, RSV-promoter, CMV-enhancer, SV40-enhancer or a globin intron in mammalian and other animal cells. Beside elements which are responsible for the initiation of transcription such regulatory elements may also comprise transcription termination signals, such as the SV40-poly-A site or the tk-poly-A site, downstream of the polynucleotide. Furthermore, depending on the expression system used leader sequences capable of directing the polypeptide to a cellular compartment or secreting it into the medium may be added to the coding sequence of the polynucleotide of the invention and are well known in the art. The leader sequence(s) is (are) assembled in appropriate phase with translation, initiation and termination sequences, and preferably, a leader sequence capable of directing secretion of translated protein, or a portion thereof, into the periplasmic space or extracellular medium. Optionally, the heterologous sequence can encode a fusion protein including a C- or N-terminal identification peptide imparting desired characteristics, e.g., stabilization or simplified purification of expressed recombinant product. Preferably, the expression control sequences will be eukaryotic promoter systems in vectors capable of transforming or transfecting eukaryotic host cells, but control sequences for prokaryotic hosts may also be used. Once the vector has been incorporated into the appropriate host, the host is maintained under conditions suitable for high level expression of the nucleotide sequences, and, as desired, the collection and purification of the immunoglobulin light chains, heavy chains, light/heavy chain dimers or intact antibodies, binding fragments or other immunoglobulin forms may follow; see, Beychok, Cells of Immunoglobulin Synthesis, Academic Press, N.Y., (1979).

(83) Furthermore, the present invention relates to vectors, particularly plasmids, cosmids, viruses and bacteriophages used conventionally in genetic engineering that comprise a polynucleotide encoding the antigen or preferably a variable domain of an immunoglobulin chain of an antibody of the invention; optionally in combination with a polynucleotide of the invention that encodes the variable domain of the other immunoglobulin chain of the antibody of the invention. Preferably, said vector is an expression vector and/or a gene transfer or targeting vector. Expression vectors derived from viruses such as retroviruses, vaccinia virus, adeno-associated virus, herpes viruses, or bovine papilloma virus, may be used for delivery of the polynucleotides or vector of the invention into targeted cell population. Methods which are well known to those skilled in the art can be used to construct recombinant viral vectors; see, for example, the techniques described in Sambrook, Molecular Cloning A Laboratory Manual, Cold Spring Harbor Laboratory (1989) N.Y. and Ausubel, Current Protocols in Molecular Biology, Green Publishing Associates and Wiley Interscience, N.Y. (1994). Alternatively, the polynucleotides and vectors of the invention can be reconstituted into liposomes for delivery to target cells. The vectors containing the polynucleotides of the invention (e.g., the heavy and/or light variable domain(s) of the immunoglobulin chains encoding sequences and expression control sequences) can be transferred into the host cell by well-known methods, which vary depending on the type of cellular host. For example, calcium chloride transfection is commonly utilized for prokaryotic cells, whereas calcium phosphate treatment or electroporation may be used for other cellular hosts; see Sambrook, supra.

(84) The present invention furthermore relates to host cells transformed with a polynucleotide or

vector of the invention. Said host cell may be a prokaryotic or eukaryotic cell. The polynucleotide or vector of the invention which is present in the host cell may either be integrated into the genome of the host cell or it may be maintained extrachromosomally. The host cell can be any prokaryotic or eukaryotic cell, such as a bacterial, insect, fungal, plant, animal or human cell. Preferred fungal cells are, for example, those of the genus *Saccharomyces*, in particular those of the species *S. cerevisiae*. The term "prokaryotic" is meant to include all bacteria which can be transformed or transfected with a DNA or RNA molecules for the expression of an antibody of the invention or the corresponding immunoglobulin chains. Prokaryotic hosts may include gram negative as well as gram positive bacteria such as, for example, *E. coli, S. typhimurium, Serratia marcescens* and *Bacillus subtilis.* The term "eukaryotic" is meant to include yeast, higher plant, insect and preferably mammalian cells, most preferably HEK293, NSO and CHO cells. Depending upon the host employed in a recombinant production procedure, the antibodies or immunoglobulin chains encoded by the polynucleotide of the present invention may be glycosylated or may be non-glycosylated. Antibodies of the invention or the corresponding immunoglobulin chains may also include an initial methionine amino acid residue. A polynucleotide of the invention can be used to transform or transfect the host using any of the techniques commonly known to those of ordinary skill in the art. Furthermore, methods for preparing fused, operably linked genes and expressing them in, e.g., mammalian cells and bacteria are well-known in the art (Sambrook, Molecular Cloning: A Laboratory Manual, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY, 1989). The genetic constructs and methods described therein can be utilized for expression of the antibody of the invention or the corresponding immunoglobulin chains in eukaryotic or prokaryotic hosts. In general, expression vectors containing promoter sequences which facilitate the efficient transcription of the inserted polynucleotide are used in connection with the host. The expression vector typically contains an origin of replication, a promoter, and a terminator, as well as specific genes which are capable of providing phenotypic selection of the transformed cells. Suitable source cells for the DNA sequences and host cells for immunoglobulin expression and secretion can be obtained from a number of sources, such as the American Type Culture Collection ("Catalogue of Cell Lines and Hybridomas," Fifth edition (1985) Rockville, Maryland, U.S.A., which is incorporated herein by reference). Furthermore, transgenic animals, preferably mammals, comprising cells of the invention may be used for the large scale production of the antibody of the invention.

(85) A further aspect refers to a method for preparing an anti-allergen antibody or allergen-binding fragment thereof, consisting of culturing the cell comprising a polynucleotide encoding the antibody or binding fragment thereof as described herein and isolating the antibody or allergen binding fragment thereof from the cell or culture medium of the cell.

(86) In a further embodiment, the present invention relates to a method for the production of an antibody or a binding fragment thereof, said method comprising

(87) (a) culturing a cell as described herein; and

(88) (b) isolating said antibody or binding fragment thereof from the culture.

(89) The transformed hosts can be grown in fermentors and cultured according to techniques known in the art to achieve optimal cell growth. Once expressed, the whole antibodies, their dimers, individual light and heavy chains, or other immunoglobulin forms of the present invention, can be purified according to standard procedures of the art, including ammonium sulfate precipitation, affinity columns, column chromatography, gel electrophoresis and the like; see, Scopes, "Protein Purification", Springer Verlag, N.Y. (1982). The antibody or its corresponding immunoglobulin chain(s) of the invention can then be isolated from the growth medium, cellular lysates, or cellular membrane fractions. The isolation and purification of the, e.g., recombinantly expressed antibodies or immunoglobulin chains of the invention may be by any conventional means such as, for example, preparative chromatographic separations and immunological separations such as those involving the use of monoclonal or polyclonal antibodies directed, e.g.,

against the constant region of the antibody of the invention. It will be apparent to those skilled in the art that the antibodies of the invention can be further coupled to other moieties for, e.g., drug targeting and imaging applications. Such coupling may be conducted chemically after expression of the antibody or antigen to site of attachment or the coupling product may be engineered into the antibody or antigen of the invention at the DNA level. The DNAs are then expressed in a suitable host system, and the expressed proteins are collected and renatured, if necessary.
(90) Substantially pure immunoglobulins of at least about 90 to 95% homogeneity are preferred, and 98 to 99% or more homogeneity most preferred, for pharmaceutical uses. Once purified, partially or to homogeneity as desired, the antibodies may then be used therapeutically (including extracorporally) or in developing and performing assay procedures.
(91) The present invention also involves a method for producing cells capable of expressing an antibody of the invention or its corresponding immunoglobulin chain(s) comprising genetically engineering cells with the polynucleotide or with the vector of the invention.
(92) Moreover, the invention relates to an antibody composition comprising at least two antibodies, wherein at least one of the antibodies is selected from the antibodies as defined herein. In one embodiment the antibody composition comprises at least two of the antibodies selected from the antibodies as defined herein. In one embodiment the antibody composition comprises at least three of the antibodies selected from the antibodies as defined herein.
(93) In one embodiment, the antibody composition comprises the antibodies 32B10, 37D5 and 7G6.
(94) Another aspect of the invention refers to a pharmaceutical composition comprising at least one of the compounds selected from the group consisting of antibody or binding fragment as described herein, or the antibody composition as described herein, the polynucleotide as described herein, the vector as described herein and the cell as described herein.
(95) In one embodiment, the pharmaceutical comprises the antibody or binding fragment of any as described herein or the antibody composition as described herein.
(96) In another embodiment the pharmaceutical composition further comprises an additional agent useful for treating peanut allergy. The additional agent useful for treating peanut allergy may be a β-adrenergic agonist, such as epinephrine, antihistamine, corticosteroid, anti-IgE antibody, anti-IgE antibody binding fragment, peptide vaccine and further antibodies capable of binding to a peanut allergen. Preferably, the pharmaceutical composition comprises epinephrine.
(97) Such compositions pharmaceutical comprise a therapeutically or prophylactically effective amount of an antibody or binding fragment thereof in admixture with a suitable pharmaceutical acceptable carrier, e.g., a pharmaceutically acceptable agent.
(98) Pharmaceutically acceptable agents for use in the present pharmaceutical compositions include carriers, excipients, diluents, antioxidants, preservatives, coloring, flavoring and diluting agents, emulsifying agents, suspending agents, solvents, fillers, bulking agents, buffers, delivery vehicles, tonicity agents, co-solvents, wetting agents, complexing agents, buffering agents, antimicrobials, and surfactants.
(99) The composition can be in liquid form or in a lyophilized or freeze-dried form and may include one or more lyoprotectants, excipients, surfactants, high molecular weight structural additives and/or bulking agents (see for example U.S. Pat. Nos. 6,685,940, 6,566,329, and 6,372,716).
(100) Compositions can be suitable for parenteral administration. Exemplary compositions are suitable for injection or infusion into an animal by any route available to the skilled worker, such as intraarticular, subcutaneous, intravenous, intramuscular, intraperitoneal, intracerebral (intraparenchymal), intracerebroventricular, intramuscular, intraocular, intraarterial, or intralesional routes. A parenteral formulation typically will be a sterile, pyrogen-free, isotonic aqueous solution, optionally containing pharmaceutically acceptable preservatives. Preferably the pharmaceutical composition is intended for subcutaneous, intravenous, intramuscular, intraperitoneally intranasal

and/or inhalative administration.
(101) Examples of non-aqueous solvents are propylene glycol, polyethylene glycol, vegetable oils such as olive oil, and injectable organic esters such as ethyl oleate. Aqueous carriers include water, alcoholic/aqueous solutions, emulsions or suspensions, including saline and buffered media. Parenteral vehicles include sodium chloride solution, Ringers' dextrose, dextrose and sodium chloride, lactated Ringer's, or fixed oils. Intravenous vehicles include fluid and nutrient replenishers, electrolyte replenishers, such as those based on Ringer's dextrose, and the like. Preservatives and other additives may also be present, such as, for example, anti-microbials, anti-oxidants, chelating agents, inert gases and the like. See generally, Remington's Pharmaceutical Science, 16th Ed., Mack Eds., 1980, which is incorporated herein by reference.
(102) Pharmaceutical compositions described herein can be formulated for controlled or sustained delivery in a manner that provides local concentration of the product (e.g., bolus, depot effect) and/or increased stability or half-life in a particular local environment. The compositions can include the formulation of antibodies, binding fragments, nucleic acids, or vectors of the invention with particulate preparations of polymeric compounds such as polylactic acid, polyglycolic acid, etc., as well as agents such as a biodegradable matrix, injectable microspheres, microcapsular particles, microcapsules, bioerodible particles beads, liposomes, and implantable delivery devices that provide for the controlled or sustained release of the active agent which then can be delivered as a depot injection.
(103) Both biodegradable and non-biodegradable polymeric matrices can be used to deliver compositions of the present invention, and such polymeric matrices may comprise natural or synthetic polymers. Biodegradable matrices are preferred. The period of time over which release occurs is based on selection of the polymer. Typically, release over a period ranging from between a few hours and three to twelve months is most desirable.
(104) Alternatively or additionally, the compositions can be administered locally via implantation into the affected area of a membrane, sponge, or other appropriate material on to which an antibody, binding fragment, nucleic acid, or vector of the invention has been absorbed or encapsulated. Where an implantation device is used, the device can be implanted into any suitable tissue or organ, and delivery of an antibody, binding fragment, nucleic acid, or vector of the invention can be directly through the device via bolus, or via continuous administration, or via catheter using continuous infusion.
(105) A pharmaceutical composition comprising a binding antibody or binding fragment thereof can be formulated for inhalation, such as for example, as a dry powder. Inhalation solutions also can be formulated in a liquefied propellant for aerosol delivery. In yet another formulation, solutions may be nebulized.
(106) Certain formulations containing antibodies or binding fragments thereof can be administered orally. Formulations administered in this fashion can be formulated with or without those carriers customarily used in the compounding of solid dosage forms such as tablets and capsules. For example, a capsule can be designed to release the active portion of the formulation at the point in the gastrointestinal tract when bioavailability is maximized and pre-systemic degradation is minimized. Additional agents can be included to facilitate absorption of a selective binding agent. Diluents, flavorings, low melting point waxes, vegetable oils, lubricants, suspending agents, tablet disintegrating agents, and binders also can be employed.
(107) The phrase “therapeutically or prophylactically effective amount” as used herein refers to an amount that produces the desired effect for which it is administered. The exact amount will depend on the purpose of the treatment, and will be ascertainable by one skilled in the art using known techniques (see, for example, Lloyd (2012) The Art, Science and Technology of Pharmaceutical Compounding).
(108) Other aspects of the invention refer to the antibodies or binding fragments, antibody composition, polynucleotide, vector, cell or pharmaceutical composition as described herein for use

in the treatment of allergy. The allergy may be a food allergy, such as peanut allergy.
(109) In particular, the invention refers to the antibodies or binding fragments as described herein, the antibody composition as described herein and the pharmaceutical composition as described herein for use in the treatment of food allergy, such as peanut allergy. The treatment may be prophylactic or therapeutic.
(110) Correspondingly, the application relates to methods of treating allergies, in particular food allergies, such as peanut allergy by administering the antibodies or binding fragments, antibody composition, polynucleotide, vector, cell or pharmaceutical composition according to the invention. Preferably, antibodies or binding fragments thereof, the antibody composition and the pharmaceutical composition according to the invention are administered.
(111) Further, the application relates to an antibody or binding fragment thereof, antibody composition, polynucleotide, vector, cell or pharmaceutical composition according to the invention in the manufacture of a medicament for the treatment of allergies, in particular food allergies, such as peanut allergy.
(112) Another aspect refers to a method of evaluating the capacity of an candidate antibody or binding fragment thereof to inhibit allergen binding/and/or allergen-induced activity in an human, wherein the method comprises
(113) (i) incubating the candidate antibody or binding fragment thereof, a composition comprising IgEs derived from the human and a food allergen;
(114) (ii) evaluating whether the candidate antibody inhibits allergen binding to IgEs/and/or allergen-induced activity in the composition comprising IgEs derived from the human.
(115) The term “evaluating” comprises “detecting” or “determining” which lead yes or no output and to “measuring” which gives numerical values, such as continuous data values, as output.
(116) In one embodiment the method further comprises the step of
(117) (iii) determining whether the administration of the candidate antibody is a suitable treatment for a patient suffering from food allergy based on the result of step (ii).
(118) The administration of the candidate antibody is considered as suitable treatment for a patient suffering from food allergy, if the candidate antibody inhibits allergen binding to IgEs and/or inhibits allergen-induced activity in the composition comprising IgEs derived from the human.
(119) In other words, the invention refers to an in vitro method for determining efficacy of treatment of a subject suffering from food allergy, in particular peanut allergy by treatment with an antibody or fragment thereof according to the invention, comprising determining in vitro whether the candidate antibody inhibits allergen binding to the IgEs and/or inhibits allergen-induced activity in the composition comprising IgEs derived from the human.
(120) Therefore, the present application refers to methods for determining whether a patient suffering from food allergy is responsive to a candidate antibody, comprising the steps of
(121) (i) incubating the candidate antibody or binding fragment thereof, a composition comprising IgEs derived from the patient and an allergen;
(122) (ii) evaluating whether the candidate antibody inhibits allergen binding/and/or allergen-induced activity in the composition comprising IgEs derived from the patient;
(123) (iii) determining whether the administration of the candidate antibody is a suitable treatment for the patient suffering from food allergy based on the result of step (ii).
(124) In a specific embodiment, the present application refers to a method for determining whether a patient suffering from food allergy is responsive to a candidate antibody, comprising the steps of
(125) (i) incubating the candidate antibody or binding fragment thereof, a composition comprising IgEs derived from the patient and an allergen;
(126) (ii) evaluating whether the candidate antibody inhibits allergen binding and/or allergen-induced activity in the composition comprising IgEs derived from the patient by measuring the level of activated leukocytes;
(127) (iii) determining whether the administration of the candidate antibody is a suitable treatment

for the patient suffering from food allergy based on the result of step (ii).
(128) wherein decreased levels of activated leukocytes compared to a control indicate that the antibody is capable of inhibiting allergen binding and/or allergen induced activity in the human patient;
(129) Thus, the invention refers to the antibody or binding fragment thereof according to the invention for use in the treatment of food allergy, in particular peanut allergy, of a human wherein the antibody is administered if the antibody or binding fragment thereof inhibits allergen binding to the IgEs and/or inhibits allergen-induced activity in the composition comprising IgEs derived from the human. The skilled person understands that the test whether or not the antibody or binding fragment thereof inhibits allergen binding to the IgEs and/or inhibits allergen-induced activity in the composition comprising IgEs derived from the human is carried out in vitro.
(130) The skilled person is aware of methods for testing the capability of the antibody or binding fragment thereof to compete for the allergen (as described for example in Uermösi et al., Mechanism of allergen-specific desensitization, Allergy, 2010) For example a competition ELISA assay as described in example 4 could be used. In brief, allergen coated plates are incubated with increasing concentration of the antibodies to be tested, e.g. for 1 h at room temperature. Subsequently plates are incubated with the composition comprising the IgEs derived from the human. Total IgE levels can be detected using an appropriate antibody that specifically binds to IgE.
(131) Thus, one embodiment refers to a method of evaluating the capability of a candidate antibody or binding fragment thereof to inhibit allergen binding, comprising the following steps:
(132) (i) incubating the candidate antibody or binding fragment thereof together with a composition comprising IgEs derived from the human and a food allergen;
(133) (ii) measuring the level of IgE bound to antigen;
(134) wherein decreased levels of IgE bound to antigen compared to a control indicate that the antibody is capable of inhibiting allergen binding.
(135) Thereby, decreased levels of IgE bound to antigen compared to the control indicated that the antibody is suitable for the treatment of a patient suffering from food allergy.
(136) Typically, the control comprises IgEs derived from the human and a food allergen but does not contain the candidate antibody. The candidate antibody is an antibody or binding fragment thereof as defined herein.
(137) The skilled person is aware of methods for testing the capability of the antibody or binding fragment thereof to bind to the allergen (see example 2). For example a binding ELISA assay could be used. In brief, allergen coated plates are incubated with increasing concentration of the antibodies to be tested, e.g. for 1 h at room temperature. An appropriate detection antibody is used to measure concentrations/levels of anti-allergen antibody.
(138) The determination whether the antibodies or fragments thereof of the invention inhibit allergen-induced activity in the composition comprising IgEs derived from the human will be ascertainable by one skilled in the art using known techniques (see, for example Hausmann et al., Robust expression of CCR3 as single basophil marker', Allergy, 2011).
(139) For example the Basophil activation test could be used. In brief the allergen was preincubated with one or serial dilutions of the candidate antibody. Afterwards, 100 μl of whole blood of the patient were stimulated with the allergen or stimulation buffer (negative control) in the presence/absence of the candidate antibody or isotype control for 30 minutes at 37° C., 5% CO_2. Cells are simultaneously stained with anti-CCR3-APC, anti-CD203c-PE and anti-CD63-FITC (Biolegend, San Diego, CA, USA). Basophils are gated as SSClow, CCR3high lymphocytes. At least 500 basophils are acquired using e.g. a FACSCalibur (Becton Dickinson AG, Allschwil, Switzerland). Activation of basophils is quantified using % of CD63+ CCR3+ basophils.
(140) The term “isotype control” as used herein refers to an antibody of the same isotype and having the same constant region as the candidate antibody but being not specific for the allergen.

(141) Thus, one embodiment refers to a method of evaluating the capability of a candidate antibody or binding fragment thereof to inhibit allergen binding and/or allergen-induced activity in a human, comprising the following steps:
(142) (i) incubating the candidate antibody or binding fragment thereof together with basophils derived from the human and a food allergen;
(143) (ii) measuring the level of activated basophils;
(144) wherein decreased levels of activated basophils compared to a control indicate that the antibody is capable of inhibiting allergen binding and/or allergen induced activity in a human.
(145) Thereby, decreased levels of activated basophils compared to the control indicated that the antibody is suitable for the treatment of a patient suffering from food allergy.
(146) Typically, the control comprises basophils derived from the human and a food allergen but does not contain the candidate antibody.
(147) Preferably the candidate antibody is an antibody or binding fragment thereof according to the invention.
(148) The skilled person understands that the step of determining whether candidate antibody inhibits allergen binding to the IgEs and/or inhibits allergen-induced activity in the composition comprising IgEs also encompasses that several candidate antibodies or fragments thereof are tested on the same sample together (are pooled) in a first step, and if the result is positive (i.e. that the allergen binding and/or allergen-induced activity is inhibited), then in a second step the candidate antibodies or fragments thereof are tested individually. Alternatively, the candidate antibodies are already tested individually in the first step.
(149) Preferably, in step (i) the composition comprising IgEs from the human, in particular the allergic patient comprises basophils. Typically, the basophils are derived from the allergicpatient. Alternatively, the basophils are donor-derived IgE-stripped basophils.
(150) “IgE-stripped basophils” may be produced by incubating basophils with lactic acid, in order to strip off the IgEs from the donor. In order to furnish the stripped basophils with patient derived IgEs, the IgE-stripped basophils may be incubated with plasma from an allergic patient.
(151) For the identification of basophils, in particular activated basophils, cell sorting such as fluorescent activated cell sorting could be used. In one embodiment, basophils are identified based on the low side-scattered light (SSC) and the expression of CCR3. In addition, activated basophils are identified by measuring surface expression of CD63 and/or surface expression of CD203c. Typically, in activated basophils the surface expression of CD63 and/or CD203c is high.
(152) The composition comprising IgEs derived from the human may be plasma, sera, blood, saliva, peripheral blood mononuclear cell (PBMC), leukocytes, basophils or IgEs stripped from basophils.
(153) In a specific embodiment, evaluating the capability of a candidate antibody or binding fragment thereof to inhibit allergen binding and/or allergen-induced activity is accomplished by an assay measuring an inflammatory mediator released from the basophil, preferably leukotriene. This assay is advantageous, since the measurement of the leukotriene released from the basophils can be detected by ELISA and therefore can be easily carried out in high-throughput format. The leukotriene release assay allows to evaluate the capability of a candidate antibody or binding fragment thereof to inhibit allergen binding and/or allergen-induced activity with high sensitivity, reliability and cost effectiveness.
(154) In assays which determine the basophil activation by measuring the secretion of the inflammatory mediator, in particular leukotriene, from the basophils, the IgE containing composition may be isolated leukocytes. The leukocytes may be either derived from an allergic patient or may be donor-derived leukocytes, from which the IgEs from the donor are stripped-off and replaced by IgEs derived from an allergic patient.
(155) The skilled person understands that leukotrienes may be secreted also by mast cells, eosinophils, neutrophils. However, in this experimental set-up, the leukotriene secretion from these

cell types is neglectable in comparison to the secretion of leukotrienes by basophils.
(156) The leukotriene assay is described in detail in the examples. In brief, leukocytes may be isolated by dextran sedimentation from whole blood of allergic patients. Titrations of peanut extract are preincubated with purified antibody samples. Afterwards, leukocytes are stimulated with titrated peanut extract or stimulation buffer (negative control) in the presence/absence of different anti-peanut allergen antibodies or isotype control. Activation of leukocytes, in particular, basophils is quantified using CAST®ELISA (Bühlmann, Schönenbuch, Switzerland).
(157) "Leukotriene" as used herein refers to a family of eicosanoid inflammatory mediators. In particular, the term "leukotriene" refers to sulfidoleukotrienes, also termed cysteinyl leukotrienes, such as LTC4 and its metabolites LTD4 and LTE4.
(158) Leukotriene is released by activated basophils. Thus decreased leukotriene levels detected for the candidate antibody or binding fragment thereof compared to a control sample indicate that the antibody is capable of inhibiting allergen binding and/or allergen induced activity in a human.
(159) Leukotriene is released from the basophils in a dose dependent manner. Thus, it is particularly suitable for highly sensitive assays.
(160) Therefore, the methods described above can be used to compare the different candidate antibodies for the capability to treat peanut allergy, wherein a first candidate antibody leading to a lower activation of the basophils, e.g. indicated by a lower secretion of leukotriene in comparison to a second candidate antibody, indicates that the first candidate antibody is particularly suited for treating peanut allergy.
(161) Accordingly, one embodiment refers to a method of evaluating the capability of a candidate antibody or binding fragment thereof to inhibit allergen binding and/or allergen-induced activity in a human, comprising the following steps:
(162) (i) incubating the candidate antibody or binding fragment thereof together with a composition comprising IgEs derived from the human and a food allergen;
(163) (ii) evaluating whether the candidate antibody inhibits allergen binding/and/or allergen-induced activity in the composition comprising IgEs derived from the patient by measuring the leukotriene secretion;
(164) wherein decreased levels of leukotriene secretion compared to a control indicate that the antibody is capable of inhibiting allergen binding and/or allergen induced activity in a human.
(165) In another preferred embodiment, the present application refers to a method for determining whether a patient suffering from food allergy is responsive to a candidate antibody, comprising the steps of
(166) (i) incubating the candidate antibody or binding fragment thereof, a composition comprising IgEs derived from the patient and an allergen;
(167) (ii) evaluating whether the candidate antibody inhibits allergen binding/and/or allergen-induced activity in the composition comprising IgEs derived from the patient by measuring the leukotriene secretion;
(168) (iii) determining whether the administration of the candidate antibody is a suitable treatment for the patient suffering from food allergy based on the result of step (ii).
(169) wherein decreased levels of leukotriene secretion compared to a control indicate that the antibody is capable of inhibiting allergen binding and/or allergen induced activity in the human patient;
(170) Another aspect of the invention refers to a method of detecting or quantifying whether an allergen is present in a sample comprising the following steps:
(171) i) incubation of the sample with an antibody of the invention as described herein or with an antibody composition of the invention as described herein,
(172) ii) detecting the antibody which is bound to allergen in the sample.
(173) The antibody may be detectably labeled. Alternatively, the antibody is used with a second antibody that is detectably labeled. Preferably, the antibody is unlabeled and used in combination

with a second antibody that is detectably labeled. The detectable label may be selected from the group consisting of an enzyme, a radioisotope, a fluorophore, a peptide and a heavy metal.
(174) Labeling agents can be coupled either directly or indirectly to the antibodies or antigens of the invention. One example of indirect coupling is by use of a spacer moiety. Furthermore, the antibodies of the present invention can comprise a further domain, said domain being linked by covalent or non-covalent bonds. The linkage can be based on genetic fusion according to the methods known in the art and described above or can be performed by, e.g., chemical cross-linking as described in, e.g., international application WO94/04686. The additional domain present in the fusion protein comprising the antibody of the invention may preferably be linked by a flexible linker, advantageously a polypeptide linker, wherein said polypeptide linker comprises plural, hydrophilic, peptide-bonded amino acids of a length sufficient to span the distance between the C-terminal end of said further domain and the N-terminal end of the antibody of the invention or vice versa. The therapeutically or diagnostically active agent can be coupled to the antibody of the invention or an antigen-binding fragment thereof by various means. This includes, for example, single-chain fusion proteins comprising the variable regions of the antibody of the invention coupled by covalent methods, such as peptide linkages, to the therapeutically or diagnostically active agent. Further examples include molecules which comprise at least an antigen-binding fragment coupled to additional molecules covalently or non-covalently include those in the following non-limiting illustrative list.

EXPERIMENTS

Examples

(175) The following examples further illustrate the invention, but should not be construed to limit the scope of the invention in any way. Detailed descriptions of conventional methods, such as those employed herein can be found in the cited literature. The practice of the present invention will employ, unless otherwise indicated, conventional techniques of cell biology, cell culture, molecular biology, transgenic biology, microbiology, recombinant DNA, and immunology, which are within the skill of the art.

Material and Methods

Patients Selection and Disease Associations of the Invention

(176) As starting material for the cloning of fully human antibodies, human lymphocytes were obtained from peripheral blood of 84 voluntary allergic patients. All patients gave their written informed consent and the study has been approved by the ethical Committee of Zurich. Clinical reactivity was defined based on anamnesis of signs of allergic reactions (e.g. oral allergy syndrome, angioedema, dyspnea, nausea, emesis, diarrhea, shock) and/or positive skin prick test or presence of serum IgE antibodies (ImmunoCAP, Phadia, Uppsala, Sweden).

Isolation of Memory B Cells Reactive to Peanut Allergens

(177) Antibodies specific to major peanut allergens were isolated by molecular cloning of immunoglobulin genes obtained from single-cell sorted cells derived from short term oligoclonal cultures of activated memory B cells producing the antibodies of interest.

Isolation of PBMC and Memory B Cell Culture

(178) Peripheral blood mononuclear cells (PBMCs) were isolated from heparinized peripheral blood using Lympholyte H according to manufacturer's instruction (Cedarlane, Burlington, Ontario, Canada) and cryopreserved prior to use.
(179) Cells were stained on ice with monoclonal antibodies phycoerythrin-conjugated anti-human IgD and IgA, APC-conjugated mAbs anti-human IgM, CD3, CD56, CD8, and FITC-conjugated mAb anti-human CD22 (Becton Dickinson, Basel, Switzerland). Cells sorting was carried out using a MoFlo XDP cell sorter (Beckman Coulter, Krefeld, Germany). CD22 positive and IgM, IgD, IgA negative B cells were seeded at 5-10 cells per well on irradiated CD40L-expressing feeder cells stimulated with a cytokine cocktail, as described in Huang et al., Isolation of human monoclonal antibodies from peripheral blood cells', Nature Protocols, 2012.

(180) After 10-14 days of stimulation, culture supernatants were screened for the presence of IgG or IgE antibodies specific for the target of interest (e.g. Ara h 1, Ara h 2, Ara h 3, Ara h 6, peanut extract). The screening process comprised for binding on recombinant or naturally extracted peanut allergens of the particular molecule of interest (e.g. by ELISA). Detection of peanut-specific IgE or IgG antibodies is performed using anti-human HRP-conjugated Fc-epsilon-specific secondary antibody (Abcam, Cambridge, UK) or anti-human HRP-conjugated goat Fc-gamma-specific antibody (Jackson ImmunoResearch, West Grove, PA, USA) followed by measurement of the HRP activity using a tetramethylbenzidine substrate solution (TMB, Sigma-Aldrich Chemie GmbH, Buchs, Switzerland). Subsequently the antibody for which binding is detected or the cell producing said antibody is isolated.

Molecular Cloning of Human Antibodies Specific to Peanut Allergens

(181) Molecular cloning of human antibodies specific to peanut allergens is carried out according to Huang et al., Isolation of human monoclonal antibodies from peripheral blood cells', Nature Protocols, 2012. In particular, single cells obtained from peanut allergen-reactive memory B cell cultures are sorted into a 96 well PCR plate, containing reverse-transcription buffer (Invitrogen, Carlsbad, Calif., United States). cDNA preparation is done using Random Hexamer Primer (Invitrogen, Carlsbad, CA, United States) according to the supplier's protocol. Immunoglobulin heavy and light chain variable regions (V.sub.H and V.sub.L, respectively) are PCR amplified according to standard protocols (Wardemann et al, Science 301, 2003, 1374-1377). To increase the PCR efficiency, a nested PCR is performed. The first round PCR is performed with primers specific for the IgG and IgE constant region and primer mixes specific to all leader sequences from V.sub.H and V.sub.L families (Wardemann et al, Science 301, 2003, 1374-1377). Subsequently, a nested PCR is performed using primer mixes specific to the 5′ region of framework 1 of V.sub.H and V.sub.L families (V-region) and the immunoglobulin J-regions. Sequence analysis is carried out to identify individual antibody clones present in the selected B cell culture. Afterwards, the V.sub.H and V.sub.L of each unique antibody clone are cloned into expression vectors providing the constant regions of human IgG1, human Ig-Kappa or human Ig-Lambda. Upon co-transfection of the Ig-heavy- and light expression vectors into HEK293T cells, the respective antibody clones are produced. Identification of the antibody clone presumably responsible for the anti-peanut reactivity of the parental B cell culture is performed upon re-screening of the recombinant antibodies for reactivity to peanut allergens in an ELISA.

(182) In order to identify and to correct primer encoded sequence mismatches in the Ig-variable regions, an additional PCR amplification on original cDNA of the reactive clone is performed using a semi-nested PCR. Therefore, primer mixes specific for the Ig heavy and light chain leader sequences (5′-primers) and two primer pairs specific for a conserved region of the Ig heavy and light chain constant regions (3′-primers) are used. PCR products are cloned into pCR™2.1-TOPO® vector (Invitrogen, Carlsbad, CA, United States) and subsequently sequenced using standard primers. Alternatively, PCR products are directly subjected to sequencing using internal primers specific to conserved regions of the constant domains of heavy and light chains. Sequence determination and annotation of the complete antibody is carried out and this information is used to design specific primers for the cloning of the authentic human V.sub.H and V.sub.L sequence into antibody (IgG) expression vectors. This approach also allows the identification of Ig isotype (and subclass) of each isolated, reactive monoclonal antibody. These V.sub.H and V.sub.L sequences are then used for the production of recombinant antibodies which are subsequently characterized in more detail.

(183) The sequences of the antibodies are set out in Table 1.

(184) TABLE-US-00001 TABLE 1 Antibody sequences. Underlined, bold nucleotides or amino acids indicate the CDRs in the sequence of the variable domain. Italic nucleotides or amino acids indicate sequences which have not been sequenced but obtainedfrom database. Nucleotide and amino acid sequences of

variable heavy (VH) and variable light (VL), constant heavy (CH) and constant light (CL) Seq. Antibody chains. ID 37D5 VH
gaggaacagctggtggagtctgggggacgtttattacggcctggggagtccctgagcctctcctgt 12 DNA
gcagccgttggattcaacttcggt**gattttggcatggcc**tgggtccgccaacctccagggaaggg
gctggagtgggtcgct**ggcatcaattggagtggccatagtacaggttttgtagactccatgaa**
gggtcgactcaccatctccagagacagcgccaagagttccctgtttctgcaaatgaacagtctgcg
aggcgaggacacggccgtgtattactgtgcgaga**gtcgggagactttgtagtggagatatttgc**
gactcaatgggtgcttttgatctgtggggccaggggacaatggtcaccgtctcttca 37D5 VH
EEQLVESGGRLLRPGESLSLSCAAVGFNFG**DFGMA**WVRQPPG 8 amino acid
KGLEWVA**GINWSGHSTGFVDSMKG**RLTISRDSAKSSLFLQM
NSLRGEDTAVYYCAR**VGRLCSGDICDSMGAFDL**WGQGTMV TVSS 37D5 VL
gacatccagatgacccagtctccttccaccctgtctgcatctataggagacagagtcaccatcactt 11 kappa-type
gt**cgggccagtcagaccattgacaactggttggcc**tggtatcaacagagaccagggaaagcc DNA
cctaaactcctgatctatc**aggcgtctagtctacaaagt**ggggtctcatcaaggttcagaggcagt
ggatctggcacagaattcactctcaccatcaccagcctgcagcctgatgactttgctacttattattgt
cagaagtctaatggctattctcgtactttcggccaggggaccaaggtggagatcaaa 37D5 VL
DIQMTQSPSTLSASIGDRVTITC**RASQTIDNWLA**WYQQRPGK 7 kappa-type
APKLLIY**QASSLQS**GVSSRFRGSGSGTEFTLTITSLQPDDFATY amino acid
YC**QKSNGYSRT**FGQGTKVEIK 37D5 CH
gcctccaccaagggcccatcggtcttccccctggcaccctcctccaagagcacctctgggggcac 14 IgG1
agcggccctgggctgcctggtcaaggactacttccccgaaccggtgacggtgtcgtggaactca subclass
ggcgccctgaccagcggcgtgcacaccttcccggctgtcctacagtcctcaggactctactccctc DNA
agcagcgtggtgaccgtgccctccagcagcttgggcacccagacctacatctgcaacgtgaatca
caagcccagcaacaccaaggtggacaagaaagttgagcccaaatcttgtgacaaaactcacacat
gcccaccgtgcccagcacctgaactcctggggggaccgtcagtcttcctcttccccccaaaaccc
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aagaccctgaggtcaagttcaactggtacgtggacggcgtggaggtgcataatgccaagacaaa
gccgcgggaggagcagtacaacagcacgtaccgtgtggtcagcgtcctcaccgtcctgcaccag
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cccgggatgagctgaccaagaaccaggtcagcctgacctgcctggtcaaaggcttctatcccagc
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cgtgctggactccgacggctccttcttcctctacagcaagctcaccgtggacaagagcaggtggc
agcaggggaacgtcttctcatgctccgtgatgcatgaggctctgcacaaccactacacgcagaag agcctctccctgtctccgggtaaatga
37D5 CH ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS 10 IgG1
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVN subclass
HKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPK amino acid
PKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKAL
PAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKS
RWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK 37D5 CL
cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgagcagttgaaatctggaactgc 13 kappa-type
ctctgttgtgtgcctgctgaataacttctatcccagagaggccaaagtacagtggaaggtggataac DNA
gccctccaatcgggtaactcccaggagagtgtcacagagcaggacagcaaggacagcacctac
agcctcagcagcaccctgacgctgagcaaagcagactacgagaaacacaaagtctacgcctgc
gaagtcacccatcagggcctgagctcgcccgtcacaaagagcttcaacaggggagagtgttag 37D5 CL
RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK 9 kappa-type
VDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKV amino acid
YACEVTHQGLSSPVTKSFNRGEC 32B10 VH

caggtgcagctggtgcagtctggggctgaggtgaagaagcctggggcctcagtgaaggtctcct 26 DNA
gcagggcttctggatacatcttccgc**aactttgatatcaact**gggtgcgacaggccactggacaa
gggcttgagtggatggga**tggatgagccctaagagtggtgacaccggctatgctcagaagtt**
ccagggcagggtcaccatgaccagggacacctccataaacacagcctacatggaactgagcag
cctgacatctgaggattcggccgtctattactgtgcgaga**ggtgtcgacgggaccaac**tggggc cagggaacccgggtcaccgtctcctca
32B10 VH QVQLVQSGAEVKKPGASVKVSCRASGYIFR**NFDIN**WVRQAT 22 amino acid
GQGLEWMG**WMSPKSGDTGYAQKFQG**RVTMTRDTSINTAY
MELSSLTSEDSAVYYCAR**GVDGTN**WGQGTRVTVSS 32B10 VL
gacatcgtgatgacccagtctccagactccctgggtgtgtctctgggcgagagggccaccatcag 25 kappa-type
ctgc**aagtccagccagagtattttagataactccaacaataagaacttcatagct**tggttcca DNA
gcagaaaccaggacagcccccctaagctgctcatttac**tgggcatctgcccgggaatcc**ggggtc
cctgaccgattcagtggcagcgggtctgggacagaattcactctcaccatcaacagcctgcaggc
tgaagatgtggcagtttattactgt**taccaatactattctactcctcacact**tttggccaggggacc aagctggatctcaga 32B10 VL
DIVMTQSPDSLGVSLGERATISC**KSSQSILDNSNNKNFIA**WFQ 21 kappa-type
QKPGQPPKLLIY**WASARES**GVPDRFSGSGSGTEFTLTINSLQA amino acid
EDVAVYYC**YQYYSTPHT**FGQGTKLDLR 32B10 CH
gcttccaccaagggcccatccgtcttccccctggcgccctgctccaggagcacctccgagagcac 28 IgG2
agccgccctgggctgcctggtcaaggactacttccccgaaccggtgacggtgtcgtggaactcag subclass
gcgctctgaccagcggcgtgcacaccttcccggctgtcctacagtcctcaggactctactccctca DNA
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agcaggggaacgtcttctcatgctccgtgatgcatgaggctctgcacaaccactacacgcaga
agagcctctccctgtctccgggtaaatga 32B10 CH
ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNS 24 IgG2
GALTSGVHTFPAVLQSSGLYSLSSVVTVTSSNFGTQTYTCNVD subclass
HKPSNTKVDKTVERKCCVECPPCPAPPVAGPSVFLFPPKPKDT amino acid
LMISRTPEVTCVVVDVSHEDPEVQFNWYVDGVEVHNA*KTKP*
REEQFNSTFRWSVLTWHQDWLNGKEYKCKVSNKGLPAPIEKTI
SKTKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEW
ESNGQPENNYKATPPMLDSDGSFFLYSKLTVDKSRWQQGNVFSC
SVMHEALHNHYTQKSLSLSPGK 32B10 CL
cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgagcagttgaaatctggaactgc 27 kappa-type
ctctgttgtgtgcctgctgaataacttctatcccagagaggccaaagtacagtggaaggtggataac DNA
gccctccaatcgggtaactcccaggagagtgtcacagagcaggacagcaaggacagcacctac
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RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK 23 kappa-type
VDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKV amino acid
YACEVTHQGLSSPVTKSFNRGEC 2F8 VH
gaagatgtccagtgtgaggtgcagctggtggagtctgggggaggcttggtcaagccggggggg 40 DNA
tcgctgagactctcctgtgcagcgtctggattcatcttcagc**gattataacatgaat**tgggtccgcc

aggctccagggaaggggctggagtgggtttca**<u>tccattactagaagtagtaggaccatttact</u>**
<u>acgcagactctgtgaagggc</u>cgattcaccatatccagagacaatgccaagaactcactgcatct
gcaaatgaacagtctcagagacgcggacacggctgtgtattattgtgcgaga**<u>gaggatttcgatg</u>**
<u>tttcgactggcccctactacatggacgtc</u>tggggcaacgggaccacggtcatcgtctcctca 2F8 VH
EVQLVESGGGLVKPGGSLRLSCAASGFIFS**<u>DYNMN</u>**WVRQAP 36 amino acid
GKGLEWVS**<u>SITRSSRTIYYADSVKG</u>**RFTISRDNAKNSLHLQM
NSLRDADTAVYYCAR**<u>EDFDVSTGPYYMDV</u>**WGNGTTVIVSS 2F8 VL
gaaattgtgttgacgcagtctccaggcaccctgtctttgtctccaggggaaagagccaccctctcct 39 kappa-type
gc**<u>agggccagtcagagtgttagcaacatgttcttagtc</u>**tggtatcagcagaaacctggccagg DNA
ctcccaggctcctcatgtat**<u>ggtgcatctaccagggccact</u>**gacatcccagacaggttcagtggc
agtgggtctgggacagacttcactctcaccatcagcagactggagcctgaagattttgcagtgtatt
actgt**<u>cagcagaatggtaactcaccatacact</u>**tttggccaggggaccaagctggagatcaaa 2F8 VL
EIVLTQSPGTLSLSPGERATLSC**<u>RASQSVSNMFLV</u>**WYQQKPG 35 kappa-type
QAPRLLMY**<u>GASTRAT</u>**DIPDRFSGSGSGTDFTLTISRLEPEDFA amino acid
VYYC**<u>QQNGNSPYT</u>**FGQGTKLEIK 2F8 CH
gcctccaccaagggcccatcggtcttccccctggcaccctcctccaagagcacctctgggggcac 42 IgG1
agcggccctgggctgcctggtcaaggactacttccccgaaccggtgacggtgtcgtggaactca subclass
ggcgccctgaccagcggcgtgcacaccttcccggctgtcctacagtcctcaggactctactccctc DNA
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2F8 CH ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS 38 IgG1
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVN subclass
HKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPK amino acid
PKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNA
KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKAL
PAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKG
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SRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK 2F8 CL
cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgagcagttgaaatctggaactgc 41 kappa-type
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gccctccaatcgggtaactcccaggagagtgtcacagagcaggacagcaaggacagcacctac
agcctcagcagcaccctgacgctgagcaaagcagactacgagaaacacaaagtctacgcctgc
gaagtcacccatcagggcctgagctcgcccgtcacaaagagcttcaacaggggagagtgttag 2F8 CL
RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK 37 kappa-type
VDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKV amino acid
YACEVTHQGLSSPVTKSFNRGEC 7G6 VH
caggtgcagctggtggagtctgggggaggcgtggtgcagcctgggaggtccctgagactctcat 54 DNA
gtgcagcctctggcatcgccttcaat**<u>gactacactatgcac</u>**tgggtccgccggtctccagacaag
ggcctggagtgggtggca**<u>gctatatcatatggtgggactaataaatactacgcagattccgtg</u>**
<u>aagggc</u>cgattcaccatctccagagacagttccaagaacaccctgtttctgcagatggacagcct

gagagttgaggacacggctgtgtattactgtgcgaga**gattctggttatcggagtcttttgcac**tg
gggccagggaaccctggtcaccgtctcctca 7G6 VH
QVQLVESGGGVVQPGRSLRLSCAASGIAFN**DYTMH**WVRRSP 50 amino acid
DKGLEWVA**AISYGGTNKYYADSVKG**RFTISRDSSKNTLFLQ
MDSLRVEDTAVYYCAR**DSGYRSLLH**WGQGTLVTVSS 7G6 VL
gagattgtgttgactcagtctccactctccctgcccgtcacccctggtgagccggcctccatctcctg 53 kappa-type
c**aggtcgagtcagagcctcgtgcatagaaatggatacaactatttagat**tggtacctgcagaa DNA
gccagggcagtctccacagctcctgatctat**atggcttctaaacgggcctcc**ggggtccctgaca
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EIVLTQSPLSLPVTPGEPASISC**RSSQSLVHRNGYNYLD**WYLQ 49 kappa-type
KPGQSPQLLIY**MASKRAS**GVPDRFSGSGSGTEFTLKISRVEAE amino acid
DVGIYYC**MQALQTWT**FGQGTKVEVN 7G6 CH
gcttccaccaagggcccatccgtcttccccctggcgccctgctccaggagcacctccgagagcac 56 IgG4
agccgccctgggctgcctggtcaaggactacttccccgaaccggtgacggtgtcgtggaactcag subclass
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atgtcttctcatgctccgtgatgcatgaggctctgcacaaccactacacacagaagagcctctccct gtctctgggtaaatga 7G6 CH
ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNS 52 IgG4
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTKTYTCNVD subclass
HKPSNTKVDKRVESKYGPPCPSCPAPEFLGGPSVFLFPPKPKD amino acid
TLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVEVHNAKTK
PREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSI
EKTISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPS
DIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSRLTVDKSRW
QEGNVFSCSVMHEALHNHYTQKSLSLSLGK 7G6 CL
cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgagcagttgaaatctggaactgc 55 kappa-type
ctctgttgtgtgcctgctgaataacttctatcccagagaggccaaagtacagtggaaggtggataac DNA
gccctccaatcgggtaactcccaggagagtgtcacagagcaggacagcaaggacagcacctac
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gaagtcacccatcagggcctgagctcgcccgtcacaaagagct*tcaacaggggagagtgttag* 7G6 CL
RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK 51 kappa-type
VDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKV amino acid
YACEVTHQGLSSPVTKS*FNRGEC* 4B2 VH
*caggtgcagctgcaggag*gcgggcccacgactggtgaagccttcacagaccctgtcagtcacc 68 DNA
tgcactgtctctggtgcctccatcacc**attggcggttactactggagt**tggatccgccagcaccca
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gtgacagccgcggacacggccatatatttctgtgcgcgg**gaggcgtgggagacgccactg**tgg
ggccagggaaccctgatcaccgtctcctcc 4B2 VH

*QVQLQE*AGPRLVKPSQTLSVTCTVSGASIT**IGGYYWS**WIRQHP 64 amino acid
GKGLEWMG**YIYSNGRTYYNPSLKS**RIAMSIDTSKNQFSLKLT
SVTAADTAIYFCAR**EAWETPL**WGQGTLITVSS 4B2 VL
gacatccagatgactcagtctccatcctccctgtctgcttctgtaggagacagagtcaccatcacttg 67 kappa-type
c**caggcgaatcaggacattgtcaactctttaaat**tggtttcaacacaaaccagggacagcccct DNA
aaagtcctgatctac**gatgcatccaaattggaaaca**ggggtcccatctaggttcagtggaagtg
ggtctgggacacattttactttcaccataagtggcctgcagcctgaagattttgcaacatatttctgtca
acaatatgagaatcttccgcacacttttggccaggggaccaagttggagatcaga 4B2 VL
DIQMTQSPSSLSASVGDRVTITC**QANQDIVNSLN**WFQHKPGT 63 kappa-type
APKVLIY**DASKLET**GVPSRFSGSGSGTHFTFTISGLQPEDFATY amino acid
FC**QQYENLPHT**FGQGTKLEIR 4B2 CH
gcctccaccaagggcccatcggtcttccccctggcaccctcctccaagagcacctctgggggcac 70 IgG1
agcggccctgggctgcctggtcaaggactacttccccgaaccggtgacggtgtcgtggaactca subclass
ggcgccctgaccagcggcgtgcacaccttcccggctgtcctacagtcctcaggactctactccctc DNA
agcagcgtggtgaccgtgccctccagcagcttgggcacccagacctacatctgcaacgtgaatca
caagcccagcaacaccaaggtggacaagaaagttgagcccaaatcttgtgacaaaactcacacat
gcccaccgtgcccagcacctgaactcctggggggaccgtcagtcttcctcttccccccaaaaccc
aaggacaccct*catgatctcccggacccctgaggtcacatgcgtggtggtggacgtgagccac*
gaagaccctgaggtcaagttcaactggtacgtggacggcgtggaggtgcataatgccaagac
aaagccgcgggaggagcagtacaacagcacgtaccgtgtggtcagcgtcctcaccgtcctgc
accaggactggctgaatggcaaggagtacaagtgcaaggtctccaacaaagccctcccagc
ccccatcgagaaaaccatctccaaagccaaagggcagccccgagaaccacaggtgtacac
cctgcccccatcccgggatgagctgaccaagaaccaggtcagcctgacctgcctggtcaaag
gcttctatcccagcgacatcgccgtggagtgggagagcaatgggcagccggagaacaacta
caagaccacgcctcccgtgctggactccgacggctccttcttcctctacagcaagctcaccgtg
gacaagagcaggtggcagcaggggaacgtcttctcatgctccgtgatgcatgaggctctgcac
aaccactacacgcagaagagcctctccctgtctccgggtaaatga 4B2 CH
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS 66 IgG1
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVN subclass
HKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPK amino acid
PKDTL*MISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTK*
PREEQYNSTYRWSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEW
ESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCS
VMHEALHNHYTQKSLSLSPGK 4B2 CL
cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgagcagttgaaatctggaactgc 69 kappa-type
ctctgttgtgtgcctgctgaataacttctatcccagagaggccaaagtacagtggaaggtggataac DNA
gccctccaatcgggtaactcccaggagagtgtcacagagcaggacagcaaggacagcacctac
agcctcagcagcaccctgacgctgagcaaagcagactacgagaaacacaaagtctacgcctgc
gaagtcacccatcagggcctgagctcgcccgtcacaaagagcttcaacaggggagagtgttag 4B2 CL
RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWK 65 kappa-type
VDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKADYEKHKV amino acid
YACEVTHQGLSSPVTKSFNRGEC 12G10 VH
*gaggtgcagctgttggag*tctgggggacgtttattacggccgggggggtccctgagcctctcctg 80 DNA
tgtagcctctggattctacttcggt**gattttggcatgagc**tgggtccgccaggttccagggaaggg
gctggagtgggtctct**ggcattgactggagtggccgtagtacaggttatgtagactccatgaa**
gggccgactcaccatctccagagacaacgacaagagttccctgtatttgcaaatgaacgatctgcg
cggcgaggacacggccgtctattactgtgcgagg**gtcgggagactctgtagtggtgattcttgc**
gactcaatgggcgctttttgacctgtggggccaggggac*aatggtcaccgtctcttca* 12G10 VH
*EVQLLE*SGGRLLRPGGSLSLSCVASGFYFG**DFGMS**WVRQVPG 78 amino acid

KGLEWVS**GIDWSGRSTGYVDSMKG**RLTISRDNDKSSLYLQ
MNDLRGEDTAVYYCAR**VGRLCSGDSCDSMGAFDL**WGQGT MVTVSS 12G10 VL
*gacatccagttgacccagtct*ccttccaccctgtctgcatctataggagacagcgtcaccatcactt 79 kappa-type
gc**cgggccagtcagaatattgataactggttggcc**tggtatcaacagaaaccagggaaagccc DNA
ctagactcctgatctac**aaggcgtctagtctaggaagt**ggggtctcatcaaagttcagaggcagt
ggatttgggacagagttcactctcaccatcaccagcctgcagcctgatgactttgcaacctattattg
t**cagaagtctaatggctattctcgtact**tttggccaggg*gaccaaagtggatatcaaa* 12G10 VL
*DIQLTQSP*STLSASIGDSVTITC**RASQNIDNWLA**WYQQKPGKA 77 kappa-type
PRLLIY**KASSLGS**GVSSKFRGSGFGTEFTLTITSLQPDDFATYY amino acid
C**QKSNGYSRT**FGQG*TKVDIK*

Antibody Production and Purification

(185) Recombinant monoclonal human antibodies are expressed upon transfection of antibody-coding expression vectors into HEK293T or Chinese Hamster Ovary cells by the Polyethylenimine transfection method (PEI, Polyscience Warrington, USA). After transfection, cells are cultured in reduced serum medium (OptiMEM® I supplemented with GlutaMAX.sup.198-I, Gibco) for 3-6 days. Subsequently, supernatants are collected and IgG-antibodies are purified using protein A columns (GE HealthCare, Sweden) on a fast protein liquid chromatography device (FPLC) (ÄKTA, GE HealthCare, Sweden).

Example 1: Detection of Peanut-Specific Antibodies by ELISA Assay

(186) Enzyme linked immunosorbent assay (ELISA) was used to evaluate allergen reactivity in the sera of allergic patients. Allergens (Ara h 2, Ara h 1, Ara h 3, Ara h 6, peanut extract) were coated on plates and anti-peanut antibody levels in sera from allergic patients were detected using HRP-conjugated anti-human antibody. Altogether, sera from 84 allergic patients, were used in the assays. The following protocol describes the experimental procedures for the detection of anti-allergen antibodies by ELISA assay. Allergic patients have shown seroreactivity against several Ara h peanut allergens, implicating the suitability of these antibodies against allergic response induced by several peanut allergens.

ELISA for the Detection of Peanut-Specific Antibodies

(187) 96 well microplates (Costar®, Corning Incorporated, Corning, NY, USA) were coated with peanut allergens either extracted and purified from peanuts or recombinantly expressed. Major peanut allergens included in the example are Ara h 1, Ara h 2, Ara h 3 and Ara h 6 (Indoor Biotechnologies, Cardiff, UK). Plates were washed with PBS-Tween 0.05% and blocked 1 h at room temperature with PBS containing 5% Milk (Rapilait, Migros, Zurich, Switzerland) or 2% bovine serum albumin (BSA, Sigma-Aldrich Chemie GmbH, Buchs, Switzerland).

(188) Patient sera, B cell conditioned medium, or recombinant antibody preparations were incubated for 2 h at room temperature. Binding of human IgG4, IgE or total IgG to the antigen of interest was determined using a HRP-conjugated anti human antibody (anti-IgG-HRP from Jackson ImmunoResearch, West Grove, Pa., USA; anti-IgE-HRP from Abcam, Cambridge, UK anti-IgG4-HRP from ThermoFisher Scientific, Carlsbad, CA, USA) followed by measurement of the HRP activity using a tetramethylbenzidine substrate solution (TMB, Sigma-Aldrich Chemie GmbH, Buchs, Switzerland). For the results see FIG. **1-2**.

Example 2: EC50 ELISA Determination of the Antibodies of the Present Invention

(189) EC50 binding of exemplary anti-peanut antibodies of the present invention to peanut allergens or peanut extract, was determined by ELISA. Serial dilutions of MAbs (from 1000 ng/ml down to 0.0169 ng/ml) were incubated for 2 hours with antigen-coated plates (coating overnight at 4° C. or 1 h at 37° C. with 1 μg/ml antigen in PBS, followed by wash out and blocking with 2% BSA in PBS). The plates were subsequently washed and binding of MAbs was detected with anti-human HRP-conjugated Fc-gamma-specific secondary antibody (Jackson ImmunoResearch, West Grove, PA, USA). Concentrations of MAb resulting in half of maximal binding to respective antigens (EC50, ng/ml) were calculated using GraphPad Prism 5 software on sigmoidal dose-

response curves (variable slope, 4 parameters) obtained by plotting the log of the concentration versus OD450 nm measurements; for the results see FIG. **3** and table 2.

(190) TABLE-US-00002 TABLE 2 EC50 ELISA determination of binding of exemplary human anti-peanut allergen antibodies and murinized versions of the antibodies depicted in FIG. 3. n.b. = not binding, n.p. = experiment not performed. EC50 [ng/ml], MY006 peanut allergens antibodies peanut other allergens [ng/ml], nArah2 extract nArah1 nArah3 nArah6 rArah8 rArah9 nBetv1 rAnao3 nCora9 h12G10 6.8 8.1 242.2 33.7 17.2 n.b. n.b. n.b. n.b. n.b. hm12G10 6.7 8.5 194.7 36.8 27.8 n.p. n.p. n.p. n.p. n.p. h37D5 5.6 6.5 263.8 65.2 22.1 n.b. n.b. n.b. n.b. n.b. hm37D5 5.2 7.1 115.1 72.3 34.0 n.p. n.p. n.p. n.p. n.p. h32B10 9.2 4.8 8.8 4.9 7.0 n.b. n.b. n.b. n.b. n.b. hm32B10 9.9 n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p. h2F8 5.6. 2.8 8.5 2.8 3.4 n.b. n.b. n.b. n.b. n.b. hm2F8 8.7 n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p. h7G6 4.7 2.8 1.9 1.3 2.6 n.b. n.b. n.b. n.b. n.b. hm7G6 n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p. h4B2 6.7 n.p. 5.9 n.p. 7.0 n.b. n.b. n.b. n.b. n.b. hm4B2 10.2 n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p. n.p.

Example 3: Cross-Competition With Human-Mouse Chimeric Constructs of Anti-Peanut Antibodies

(191) As first step of mapping, differential binding of anti-peanut MAbs of the present invention to distinct antigen binding sites was examined to determine the number of different binding sites.

(192) For this purpose, MAbs were expressed either with human (hMAb) or mouse (hmMab) Fc and cross-competition experiments were carried out by coating antigen on plates and by detecting binding of human Mabs in the presence of titrated amount of hmMabs. Detection of hMAbs bound to the ligand was performed by a HRP conjugated secondary antibody directed against the Fc portion of the primary antibody (Jackson ImmunoResearch, West Grove, PA, USA).

(193) As may be seen from Table 3, exemplary anti-peanut allergens antibodies 37D5 and 12G10 of the present invention compete each other for binding of Ara h 2 but not with antibodies 4B2 and 32B10, indicating that 12G10 and 37D5 bind other sites of Ara h 2 than 4B2 and 32B10.

(194) TABLE-US-00003 TABLE 3 The differential binding of exemplary human monoclonal antibodies (hMAbs) of the present invention with the respective MAbs with mouse Fc (hmMAbs) to distinct binding sites was investigated in cross-competition experiments with titrated hmMAbs (0-12′000 ng/ml) and anti-human IgG horseradish peroxidase (HRP)-conjugated detection antibody. 37D5 32B10 12G10 4B2 37D5 competing n.c. competing n.c. 32B10 n.c. competing n.c. n.c. 12G10 competing n.c. competing n.c. 4B2 n.c. n.c. n.c. competing “n.c.”: non-competing antibodies; “competing”: competing antibodies.

Example 4: ELISA Inhibition Assay of Allergic Patient IgE Binding to Allergen

(195) Microplates were coated with 0.05m/m1 of Ara h 1, Ara h 2, Ara h 3, or Ara h 6, overnight at 4° C. After washing with PBS-Tween 0.05% and blocking with 2% BSA, plates were incubated with increasing concentrations of peanut human IgG antibodies for 1 h at room temperature. Subsequently, plates were incubated with different sera dilutions or different concentrations of murine antibody (as positive control). IgE levels were detected using anti-human HRP-conjugated Fc-epsilon-specific secondary antibody (Abcam, Cambridge, UK) followed by addition of TMB solution (TMB, Sigma, Buchs, Switzerland). Binding of murine antibodies was detected using goat anti-mouse IgG (H+L)-HRP (Jackson ImmunoResearch, West Grove, PA, USA). Results are shown in table 4.

(196) TABLE-US-00004 TABLE 4 ELISA competition assay shows inhibition of allergic patient IgE binding to allergen (Ara h 2). 37D5 and 12G10 antibody blocked the binding of allergic patients’ polyclonal IgE to Ara h 2 up to 60%. Antibody cocktail containing 32B10, 37D5, 4B2 and 12G10 blocked up to 80% of polyclonal IgE from allergic patients. 32B10 12G10 37D5 4B2 cocktail Patient A 27.76 ± 4.85 13.82 ± 5.31 27.77 ± 3.58 39.66 ± 5.95 66.21 ± 0.6 Patient B 26.66 ± 1.63 31.16 ± 5.92 29.61 ± 1.33 30.65 ± 2.48 58.75 ± 1.49 Patient C 21.00 ± 5.17 53.81 ± 3.35 55.08.42 ± 1.98 13.34 ± 5.71 64.62 ± 3.41 Patient D 8.50 ± 7.06 35.74 ± 7.96 35.51 ± 4.53 9.23 ± 3.17 71.02 ± 1.48 Patient E 29.32 ± 2.03 36.24 ± 3.25 29.30 ± 3.05 21.91 ± 5.23 73.18 ± 5.29

Patient F 23.15 ± 4.81 45.34 ± 3.10 36.25 ± 3.27 30.77 ± 0.73 75.06 ± 6.91 Patient G 32.83 ± 6.33 25.74 ± 7.00 45.32 ± 17.11 31.77 ± 2.00 80.15 ± 5.73 Healthy 0.00 0.00 2.27 ± 64.04 0.00 0.00 donor

Example 5: Basophil Activation Test. Human-Derived Anti-Peanut Monoclonal Antibodies Neutralize Natural Ara h 2 and Peanut Extract Mediated Basophil Activation

(197) Basophil activation test was performed using whole blood from allergic patients, sampled in Heparin tubes (S-Monovette® 02.1065, Sarstedt, Nümbrecht, Germany). To test the neutralizing capacity of antibodies, the optimal concentration of Ara h 2 or peanut extract was preincubated with one or serial dilutions of serum, supernatant or purified antibody samples (one or a combination of them). Afterwards, 100 μl of whole blood were stimulated with either natural Ara h 2 (10 ng/ml, Indoor Biotechnologies, Cardiff, UK) or peanut extract (10 ng/ml, Indoor Biotechnologies, Cardiff, UK) or stimulation buffer (negative control) in the presence/absence of different anti-peanut allergen antibodies or the combination thereof (MY006 cocktail) or isotype control for 30 minutes at 37° C., 5% CO2. Cells were stained simultaneously with anti-CCR3-APC, anti-CD203c-PE and anti-CD63-FITC (Biolegend, San Diego, CA, USA). Basophils were gated as SSClow, CCR3high lymphocytes. At least 500 basophils were acquired using FACSCalibur (Becton Dickinson AG, Allschwil, Switzerland). Activation of basophils was quantified using % of CD63+ CCR3+ basophils and/or median fluorescence intensity of CD203c CCR3+basophils. Data was analyzed using FlowJo Software (FlowJo, Ashland, OR, USA). Results are shown in FIG. **4**.

Example 6: Epitope Mapping of Exemplary Peanut-Specific Antibodies

(198) A competition ELISA was used to evaluate binding of exemplary anti-peanut antibodies of the present invention to linear Ara h 2 epitopes. Serial dilutions of linear Ara h 2 epitopes (100 μM down to 0.19 nM; peptides with a length of 20 amino acids and an overlap of 12 amino acids between the peptides) were pre-incubated with exemplary anti-peanut antibodies (66 pM) for 1 hour at room temperature. The pre-incubated antibody-epitope mixtures were subsequently incubated on plates coated with full length Ara h 2 (coating overnight at 4° C. with 0.05 μg/m1 Ara h 2 in PBS, followed by wash out and blocking with 2% BSA in PBS). The plates were subsequently washed and binding of MAbs to Ara h 2 was detected HRP-conjugated goat anti human Fc-gamma-specific antibody (Jackson ImmunoResearch, West Grove, PA, USA) followed by measurement of the HRP activity using a tetramethylbenzidine substrate solution (TMB, Sigma-Aldrich Chemie GmbH, Buchs, Switzerland). As may be seen from Table 5, exemplary anti-peanut allergen antibody 32B10 binds to different linear Ara h 2 epitopes than 12G10 and 37D5. 4B2,7G6 and 2F8 do not bind to any of the 20 linear epitopes suggesting binding to conformational epitopes.

(199) TABLE-US-00005 TABLE 5 Epitope binding of exemplary human monoclonal antibodies of the present invention was evaluated in a competition ELISA with serial dilutions of linear Ara h 2 peptides (100 μM down to 0.19 nM). Antibodies 12G10 and 37D5 are binding peptides 8-11, antibody 32B10 binds to peptide 4. Antibodies 4B2, 7G6 and 2F8 do not bind any of the linear Ara h 2 peptides, “n.b.”: not binding to peptide; “Binding”: binding to peptide. SEQ ID Sequence/description of pool 32B10 37D5 12G10 4B2 7G6 2F8 POOL 1 Combination of peptides 1 to 5 Binding n.b. n.b. n.b n.b. n.b. 81 Peptide 1 .sup.1AMALKLTILVALALFLLAAH.sup.20 n.b. n.b. n.b. n.b n.b. n.b. 82 Peptide 2 .sup.9ALALFLLAAHASARQQWELQ.sup.28 n.b. n.b. n.b. n.b n.b. n.b. 83 Peptide 3 .sup.17AHASARQQWELQGDRRCQSQ.sup.36 n.b. n.b. n.b. n.b n.b. n.b. 84 Peptide 4 .sup.25WELQGDRRCQSQLERANLRP.sup.44 Binding n.b. n.b. n.b n.b. n.b. 85 Peptide 5 .sup.33CQSQLERANLRPCEQHLMQK.sup.52 n.b. n.b. n.b. n.b n.b. n.b. POOL 2 Combination of peptides 6 to 10 n.b. Binding Binding n.b n.b. n.b. 86 Peptide 6 .sup.41NLRPCEQHLMQKIQRDEDSY.sup.60 n.b. n.b. n.b n.b n.b. n.b. 87 Peptide 7 .sup.49LMQKIQRDEDSYGRDPYSPS.sup.68 n.b. n.b. n.b n.b n.b. n.b. 88 Peptide 8

.sup.57EDSYGRDPYSPSQDPYSPSQ.sup.76 n.b. Binding Binding n.b n.b. n.b. 89 Peptide 9 .sup.65YSPSQDPYSPSQDPDRRDPY.sup.84 n.b. Binding Binding n.b n.b. n.b. 90 Peptide 10 .sup.73SPSQDPDRRDPYSPSPYDRR.sup.92 n.b. Binding Binding n.b n.b. n.b. POOL 3 Combination of peptides 11 to 15 n.b. Binding Binding n.b n.b. n.b. 91 Peptide 11 .sup.81RDPYSPSPYDRRGAGSSQHQ.sup.100 n.b. Binding Binding n.b n.b. n.b. 92 Peptide 12 .sup.89YDRRGAGSSQHQERCCNELN.sup.108 n.b. n.b. n.b n.b n.b. n.b. 93 Peptide 13 .sup.97SQHQERCCNELNEFENNQRC.sup.116 n.b. n.b. n.b n.b n.b. n.b. 94 Peptide 14 .sup.105NELNEFENNQRCMCEALQQI.sup.124 n.b. n.b. n.b n.b n.b. n.b. 95 Peptide 15 .sup.113NQRCMCEALQQIMENQSDRL.sup.132 n.b. n.b. n.b n.b n.b. n.b. POOL 4 Combination of peptides 16 to 20 n.b. n.b. n.b n.b n.b. n.b. 96 Peptide 16 .sup.121LQQIMENQSDRLQGRQQEQQ.sup.140 n.b. n.b. n.b n.b n.b. n.b. 97 Peptide 17 .sup.129SDRLQGRQQEQQFKRELRNL.sup.180 n.b. n.b. n.b n.b n.b. n.b. 98 Peptide 18 .sup.137RQQEQQFKRELRNLPQQCGLRA.sup.156 n.b. n.b. n.b n.b n.b. n.b. 99 Peptide 19 .sup.145LRNLPQQCGLRAPQRCDLEV.sup.164 n.b. n.b. n.b n.b n.b. n.b. 100 Peptide 20 .sup.153GLRAPQRCDLEVESGGRDRY.sup.172 n.b. n.b. n.b n.b n.b. n.b.

Example 7: Validation of Peanut Antibodies in Animals

(200) Mice C3H/HeJ, BALB/c, C57BL/6 and/or AKR/J are sensitized by intragastric (i.g.) or intraperitoneal (i.p.) administration of peanut (e.g. ground whole peanut, crude peanut extracts or whole peanut extracts) and adjuvant (e.g. cholera toxin, alum) on a weekly basis for up to 12 weeks. Control mice receive PBS with selected adjuvant. After 2-3 months of sensitization, mice are challenged by i.g. or i.p. administration of peanut (e.g. ground whole peanut, crude peanut extracts or whole peanut extracts). Treatment consists of i.p. or intravenous (i.v.) administration of different antibody doses prior challenge.

(201) At several time points during the course of the model, blood is taken for the measurement of antibodies (total and peanut specific IgE and IgG antibodies), mMCP-1 (mast cell protease 1), histamine and leukotriene levels.

(202) Anaphylactic symptoms are evaluated 30 minutes after the challenge dose utilizing a defined scoring system (see table below). Scoring of symptoms is performed in a blinded manner by two independent investigators. After euthanization, ear, spleen and blood samples are collected. Blood samples are analyzed as described above. Histological sections are prepared and stained with toluidine blue or Giemsa stain. Degranulated mast cells are toluidine blue- or Giemsa positive with 5 or more stained granules completely outside the cell. 400 mast cells are classified.

(203) Anaphylaxis Scoring System (Ref Liu et al. 2016 Review or Original Papers):

(204) TABLE-US-00006 Score Anaphylactic symptoms 0 no symptoms 1 scratching and rubbing around the nose and head 2 puffiness around the eyes and mouth, pilar erecti (erection of the hair of the skin), reduced activity and/or decreased activity with increased respiratory rate 3 wheezing, labored respiration, and cyanosis around the mouth and tail 4 slight or no activity after prodding, or tremor and convulsion 5 death

(205) Antibody effects are evaluated by comparing clinical reactions in antibody treated and control animals using the anaphylaxis scoring system (illustrated in the table above). Therapeutic efficacy of anti-peanut antibodies disclosed herein is indicated by a reduction or absence of anaphylactic symptoms in antibody treated animals compared to control animals.

(206) Alternatively or in addition, antibody effects are evaluated by comparing serum peanut specific IgE antibodies, mMCP-1 levels and/or plasma histamine and leukotriene levels in antibody treated and control animals using ELISA-based assays. Therapeutic efficacy of anti-peanut antibodies disclosed herein is indicated by a reduction or absence of serum peanut-specific IgE antibodies and/or plasma histamine in antibody treated animals relative to control animals.

(207) Alternatively or in addition, antibody effects are evaluated by comparing cytokine expression levels in challenged splenocytes derived from antibody treated and control animals.

Example 8: Leukotriene Release Assay

(208) Leukotriene release assay was performed using whole blood from allergic patients, sampled in Heparin tubes (S-Monovette® 02.1065, Sarstedt, Nümbrecht, Germany). Leukocytes were isolated by dextran sedimentation. To test the neutralizing capacity of antibodies, titrations of peanut extract (0-30 ng/ml) were preincubated with purified antibody samples (one or a combination of them, 1.2 μg/ml) for 1 h at room temperature. Afterwards, leukocytes were stimulated with titrated peanut extract (Indoor Biotechnologies, Cardiff, UK) or stimulation buffer (negative control) in the presence/absence of different anti-peanut allergen antibodies or the combination thereof (MY006 cocktail) or isotype control for 40 minutes at 37° C., 5% CO2. Supernatant was collected and activation of basophils was quantified using CAST®ELISA (Bühlmann, Schönenbuch, Switzerland). See FIG. **5** showing that human-derived anti-peanut monoclonal antibodies neutralize peanut extract mediated basophil activation.

Example 9: Passive Sensitization Leukotriene Release Assay

(209) Passive sensitization leukotriene release assay was performed using whole blood from healthy donors, sampled in Heparin tubes. Leukocytes were isolated by dextran sedimentation. Surface IgE was removed by stripping with lactic acid, pH 3.9 (8 minutes 4° C.). Leukocytes were re-sensitized by incubation with plasma from allergic patients (1 h at 37° C.). To test the neutralizing capacity of antibodies, titrations of peanut extract (0-30 ng/ml) were preincubated with purified antibody samples (one or a combination of them, 1.2 μg/ml). Afterwards, re-sensitized leukocytes were stimulated with titrated peanut extract (Indoor Biotechnologies, Cardiff, UK) or stimulation buffer (negative control) in the presence/absence of different anti-peanut allergen antibodies or the combination thereof (MY006 cocktail) or isotype control for 40 minutes at 37° C., 5% CO2. Supernatant was collected and activation of basophils was quantified using CAST®ELISA (Bühlmann, Schönenbuch, Switzerland). See FIG. **6** showing that human-derived monoclonal antibodies decrase peanut extract mediated basophil activation.

(210) The application further contains the following embodiments:

(211) Embodiment 1. Antibody or binding fragment thereof capable of binding to a food allergen.

(212) Embodiment 2. Antibody or binding fragment thereof according to embodiment 1, wherein the antibody is human-derived.

(213) Embodiment 3. Antibody or binding fragment thereof according to embodiment 1 or 2, wherein the food allergen is a peanut allergen.

(214) Embodiment 4. Antibody or binding fragment thereof according to embodiment 3, wherein the peanut allergen is selected from the group consisting of Ara h 1, Ara h 2, Ara h 3, Ara h 4, Ara h 5, Ara h 6/7, Ara h 8, Ara h 9 and Ara h 10/11.

(215) Embodiment 5. Antibody or binding fragment thereof according to embodiment 4, wherein the peanut allergen is selected from the group consisting of Ara h 1, Ara h 2, Ara h 3 and Ara h 6, or a combination thereof.

(216) Embodiment 6. Antibody or binding fragment thereof according to any one of embodiments 1 to 5, wherein the antibody is capable of binding at least one of the Ara h2 epitopes selected from the group consisting of the amino acid sequences set forth in SEQ ID NOs: 84, 88, 89, 90 and 91.

(217) Embodiment 7. Antibody or binding fragment thereof according to any one of embodiments 3 to 5, wherein the peanut allergen is of peanut origin, recombinantly expressed or is a synthetic peanut peptide.

(218) Embodiment 8. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the antibody is a monoclonal antibody and/or wherein the antibody is a recombinant antibody.

(219) Embodiment 9. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the antibody is an IgG or IgA antibody.

(220) Embodiment 10. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the variable regions, portions thereof or the CDRs are human-derived.

(221) Embodiment 11. Antibody or binding fragment thereof according to any one of the preceding

embodiments, wherein the variable regions, portions thereof or the CDRs are derived from an IgE antibody and grafted in a scaffold of an IgG or IgA antibody.

(222) Embodiment 12. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the peptide sequence of the antibody or binding fragment thereof is identical or at least 60% identical to the sequence of the antibody extracted from the human.

(223) Embodiment 13. Antibody or binding fragment thereof according to any one of embodiment 2 to 12, wherein the human is selected from the group of a human suffering from peanut allergy, a peanut-sensitized human without clinical relevant allergy, a human suffering from peanut allergy that underwent immunotherapy, a human that has outgrown peanut allergy and a human of unknown clinical history for peanut allergy.

(224) Embodiment 14. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the antibody has an EC50 of at most 270 ng/ml, preferably at most 70 ng/ml, at most 40 ng/ml, at most 25 ng/ml, at most 15 ng/ml, at most 4.9 ng/ml, at most 1.3 ng/ml for at least one of the peanuts allergens selected from the group consisting of Ara h 2, Ara h 1, Ara h 3 and Ara h 6.

(225) Embodiment 15. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the antibody has an EC50 of at most 10 ng/ml, preferably at most 7 ng/ml, more preferably at most 4.8 ng/ml, most preferably 2.8 ng/ml for peanut extract.

(226) Embodiment 16. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the antibody is capable of reducing, inhibiting or neutralizing allergen-mediated biological activity.

(227) Embodiment 17. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the antibody is capable of reducing or inhibiting the binding of an IgE antibody to the food allergen.

(228) Embodiment 18. Antibody or binding fragment thereof according to any of the preceding embodiments, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 1 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 2 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 3 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 4 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 5 or sequences at least 65% identical thereto, and a CDR3 set forth in SEQ ID No: 6 or sequences at least 65% identical thereto.

(229) Embodiment 19. Antibody or binding fragment thereof according to any one of embodiments 1 to 17, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 15 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 16 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 17 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 18 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 19 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 20 or sequences at least 65% identical thereto.

(230) Embodiment 20. Antibody or binding fragment thereof according to any one of embodiments 1 to 17, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 29 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 30 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 31 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 32 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 33 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 34 or sequences at least 65% identical thereto.

(231) Embodiment 21. Antibody or binding fragment thereof according to any one of embodiments 1 to 17, comprising a light chain variable region and/or a heavy chain variable region, wherein the

light chain variable region comprises a CDR1 set forth in SEQ ID No: 43 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 44 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 45 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 46 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 47 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 48 or sequences at least 65% identical thereto.
(232) Embodiment 22. Antibody or binding fragment thereof according to any one of embodiments 1 to 17, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 57 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 58 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 59 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 60 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 61 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 62 or sequences at least 65% identical thereto.
(233) Embodiment 23. Antibody or binding fragment thereof according to any one of embodiments 1 to 17, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 71 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 72 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 73 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 74 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 75 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 76 or sequences at least 65% identical thereto.
(234) Embodiment 24. Antibody or binding fragment thereof according to any one of embodiments 1 to 17, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 1 or sequences at least 91% identical thereto, a CDR2 set forth in SEQ ID No: 2 or sequences at least 71% identical thereto, a CDR3 set forth in SEQ ID No: 3; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 4 or sequences at least 80% identical thereto, a CDR2 set forth in SEQ ID No: 5 or sequences at least 82% identical thereto, a CDR3 set forth in SEQ ID No: 6 or a sequence at least 94% identical thereto.
(235) Embodiment 25. Antibody or binding fragment thereof according to any one of embodiments 1 to 17, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 71 or sequences at least 91% identical thereto, a CDR2 set forth in SEQ ID No: 72 or sequences at least 71% identical thereto, a CDR3 set forth in SEQ ID No: 73; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 74 or sequences at least 80% identical thereto, a CDR2 set forth in SEQ ID No: 75 or sequences at least 82% identical thereto, a CDR3 set forth in SEQ ID No: 76 or sequences at least 94% identical thereto.
(236) Embodiment 26. Antibody or binding fragment thereof according to any one of the preceding embodiments, comprising a C.sub.L and/or C.sub.H constant region comprising an amino acid sequence selected from the C.sub.L amino acid sequences SEQ ID NOS: 9, 23, 37, 51 and 65 or an amino acid sequence with at least 60% identity and an amino acid sequence selected from the C.sub.H amino acid sequences SEQ ID NOS: 10, 24, 38, 52 and 66 or an amino acid sequence with at least 60% identity.
(237) Embodiment 27. Antibody or binding fragment thereof according to any one of the preceding embodiments, wherein the antibody is selected from the group consisting of a full length antibody, a multispecific antibody, a single chain Fv fragment (scFv), a F(ab′) fragment, a F(ab′)2 fragment, a F(ab)c fragment, a single domain antibody fragment (sdAB) and a multispecific antibody fragment.
(238) Embodiment 28. Antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a

CDR1 set forth in SEQ ID No: 1 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 2 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 3 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 4 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 5 or sequences at least 65% identical thereto, and a CDR3 set forth in SEQ ID No: 6 or sequences at least 65% identical thereto.

(239) Embodiment 29. Antibody comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 15 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 16 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 17 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 18 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 19 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 20 or sequences at least 65% identical thereto.

(240) Embodiment 30. Antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 29 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 30 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 31 or sequences at least 65% identical thereto; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 32 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 33 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 34 or sequences at least 65% identical thereto.

(241) Embodiment 31. Antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 43 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 44 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 45 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 46 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 47 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 48 or sequences at least 65% identical thereto.

(242) Embodiment 32. Antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 57 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 58 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 59 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 60 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 61 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 62 or sequences at least 65% identical thereto.

(243) Embodiment 33. Antibody or binding fragment thereof, comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 71 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 72 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 73 or sequences at least 65% identical thereto; and/or wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 74 or sequences at least 65% identical thereto, a CDR2 set forth in SEQ ID No: 75 or sequences at least 65% identical thereto, a CDR3 set forth in SEQ ID No: 76 or sequences at least 65% identical thereto.

(244) Embodiment 34. Antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 1 or sequences at least 91% identical thereto, a CDR2 set forth in SEQ ID No: 2 or sequences at least 71% identical thereto, a CDR3 set forth in SEQ ID No: 3;

wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 4 or sequences at least 80% identical thereto, a CDR2 set forth in SEQ ID No: 5 or sequences at least 82% identical thereto, a CDR3 set forth in SEQ ID No: 6 or a sequence at least 94% identical thereto.
(245) Embodiment 35. Antibody or binding fragment thereof comprising a light chain variable region and/or a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID No: 71 or sequences at least 91% identical thereto, a CDR2 set forth in SEQ ID No: 72 or sequences at least 71% identical thereto, a CDR3 set forth in SEQ ID No: 73; wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID No: 74 or sequences at least 80% identical thereto, a CDR2 set forth in SEQ ID No: 75 or sequences at least 82% identical thereto, a CDR3 set forth in SEQ ID No: 76 or sequences at least 94% identical thereto.
(246) Embodiment 36. An antibody or binding fragment thereof which competes with an antibody according to any one of embodiments 1 to 35 for specific binding to the allergen.
(247) Embodiment 37. A polynucleotide encoding the antibody or binding fragment thereof according to any one of embodiments 1 to 36.
(248) Embodiment 38. A vector comprising the polynucleotide of embodiment 37.
(249) Embodiment 39. A cell comprising the polynucleotide of embodiment 37 or the vector of embodiment 38.
(250) Embodiment 40. A method for preparing an anti-allergen antibody or allergen-binding fragment thereof, consisting of culturing the cell of embodiment 39 and isolating the antibody or allergen binding fragment thereof from the cell or culture medium of the cell.
(251) Embodiment 41. Antibody composition comprising at least two antibodies, wherein at least one of the antibodies is selected from the antibodies as defined in embodiments 18 to 26 or 28 to 35.
(252) Embodiment 42. Antibody composition according to embodiment 41, wherein at least two of the antibodies are selected from the antibodies as defined in embodiments 18 to 26 or 28 to 35.
(253) Embodiment 43. Antibody composition according to embodiment 42, wherein at least three of the antibodies are selected from the antibodies as defined in embodiments 18 to 26 or 28 to 35.
(254) Embodiment 44. A pharmaceutical composition comprising at least one of the compounds selected from the group consisting of antibody or binding fragment of any one of embodiments 1 to 36, or the antibody composition according to embodiments 41 to 43, the polynucleotide of embodiment 37, the vector of embodiment 38 and the cell of embodiment 39.
(255) Embodiment 45. A pharmaceutical composition according to embodiment 44, comprising the antibody or binding fragment of any one of embodiments 1 to 36 or the antibody composition according to embodiments 41 to 43.
(256) Embodiment 46. A pharmaceutical composition according to embodiments 44 to 45, further comprising a pharmaceutical acceptable carrier.
(257) Embodiment 47. A pharmaceutical composition according to embodiments 44 to 46, wherein the pharmaceutical composition further comprises at least one additional agent useful for treating peanut allergy.
(258) Embodiment 48. A pharmaceutical composition according to embodiment 47 wherein the additional agent useful for treating peanut allergy is selected from the group β-adrenergic agonists, epinephrine, antihistamine, corticosteroid, anti-IgE antibody, anti-IgE antibody binding fragment, peptide vaccine and further antibodies capable of binding to a peanut allergen.
(259) Embodiment 49. A pharmaceutical composition according to embodiments 44 to 48, wherein the composition is intended for subcutaneous, intravenous, intramuscular, intraperitoneally intranasal and/or inhalative administration.
(260) Embodiment 50. Kit comprising (i) the pharmaceutical composition according to embodiments 44 to 46, (ii) at least one additional agent useful for treating peanut allergy selected from the group β-adrenergic agonist, antihistamine, corticosteroid, anti-IgE antibody, anti-IgE

antibody binding fragment, peptide vaccine and further antibodies capable of binding to a peanut allergen.
(261) Embodiment 51. Kit according to embodiment 50, wherein the at least one additional agent is a β-adrenergic agonist, preferably epinephrine.
(262) Embodiment 52. Antibody or binding fragment thereof according to any one of embodiments 1 to 36, antibody composition according to embodiments 41 to 43, polynucleotide according to embodiment 37, vector according to embodiment 38, cell according to embodiment 39 or pharmaceutical composition according to embodiments 44 to 49 for use in the treatment of peanut allergy.
(263) Embodiment 53. Antibody or binding fragment, antibody composition, polynucleotide, vector, cell or pharmaceutical composition for use in the treatment of peanut allergy according to embodiment 52, wherein the treatment is prophylactic or therapeutic.
(264) Embodiment 54. A method of evaluating the capacity of a candidate antibody or binding fragment thereof to inhibit allergen binding/and/or allergen-induced activity in a human,
(265) wherein the method comprises
(266) (i) incubating the candidate antibody or binding fragment thereof with a composition comprising IgEs derived from the human and a food allergen;
(267) (ii) evaluating whether the candidate antibody inhibits allergen binding to IgEs/ and/or allergen-induced activity in the composition comprising IgEs derived from the human.
(268) Embodiment 55. The method of embodiment 54, further comprising the step of
(269) (iii) determining whether the administration of the candidate antibody is a suitable treatment for a patient suffering from food allergy based on the result of step (ii).
(270) Embodiment 56. Method according to embodiments 54 and 55, wherein the candidate antibody is an antibody as defined in embodiments 1 to 36.
(271) Embodiment 57. Method according to embodiments 54 to 56, wherein in step (i) the composition comprising IgEs from the human comprises basophils.
(272) Embodiment 58. Method according to embodiments 54 to 57, wherein the basophils are derived from an allergic patient.
(273) Embodiment 59. Method according to embodiments 54 to 58, wherein the basophils are donor-derived IgE-stripped basophils.
(274) Embodiment 60. Method according to embodiments 54 to 59, wherein the expression of CD63+at the surface of the basophil and/or the expression of CD203c is measured.
(275) Embodiment 61. Method according to embodiments 54 to 60, wherein the basophils are identified based on the CCR3 expression.
(276) Embodiment 62. Method according to embodiments 54 to 59, wherein the secretion of a mediator from basophils is measured.
(277) Embodiment 63. Method according to embodiment 62, wherein the mediator is leukotriene.
(278) Embodiment 64. Method according to embodiment 63, wherein the mediator is sulfidoleukotrine.
(279) Embodiment 65. Method according to any one of embodiments 54 to 64, wherein the composition comprising IgEs derived from the human is plasma, sera, blood, saliva, peripheral blood mononuclear cell (PBMC), leukocytes, basophils or IgEs stripped from basophils.
(280) Embodiment 66. Method according to any one of embodiments 54 to 65, wherein the human is suffering from food allergy.
(281) Embodiment 67. Method according to any one of embodiments 54 to 66, wherein the human is suffering from peanut allergy.
(282) Embodiment 68. Method according to any one of embodiments 54 to 67, wherein the food allergen is a peanut allergen.
(283) Embodiment 69. Method according to embodiment 68, wherein the peanut allergen is selected from the group consisting of Ara h 1, Ara h 2, Ara h 3, Ara h 4, Ara h 5, Ara h 6/7, Ara h 8,

Ara h 9 and Ara h 10/11.
(284) Embodiment 70. Method according to embodiment 69, wherein the peanut allergen is selected from the group consisting of Ara h 1, Ara h 2, Ara h 3 and Ara h 6, or a combination thereof.
(285) Embodiment 71. Method according to any one of embodiment 68 to 70, wherein the peanut allergen is recombinantly expressed, synthetically generated or is of peanut origin.
(286) Embodiment 72. A method of detecting or quantifying whether an allergen is present in a sample comprising the following steps:
(287) i) incubation of the sample with an antibody according to any one of embodiments 1 to 36 or with an antibody composition according to embodiments 41 to 43,
(288) ii) detecting the antibody which is bound to allergen in the sample.
(289) Embodiment 73. The method of embodiment 72, wherein the antibody is detectably labeled.
(290) Embodiment 74. The method of embodiment 72, wherein the antibody is unlabeled and used in combination with a second antibody that is detectably labeled.
(291) Embodiment 75. The method of embodiment 73 or 74, wherein the detectable label is selected from the group consisting of an enzyme, a radioisotope, a fluorophore, a peptide and a heavy metal.

Claims

1. An antibody or binding fragment thereof capable of binding to a food allergen, wherein the antibody is selected from the group consisting of: a) the antibody or binding fragment thereof comprises a light chain variable region and a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID NO: 43, a CDR2 set forth in SEQ ID NO: 44, and a CDR3 set forth in SEQ ID NO: 45; and wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID NO: 46, a CDR2 set forth in SEQ ID NO: 47, and a CDR3 set forth in SEQ ID NO: 48; and b) the antibody or binding fragment thereof comprises a light chain variable region and a heavy chain variable region, wherein the light chain variable region comprises a CDR1 set forth in SEQ ID NO: 29, a CDR2 set forth in SEQ ID NO: 30, and a CDR3 set forth in SEQ ID NO: 31; and wherein the heavy chain variable region comprises a CDR1 set forth in SEQ ID NO: 32, a CDR2 set forth in SEQ ID NO: 33, a CDR3 set forth in SEQ ID NO: 34; and wherein the food allergen is a peanut allergen.
2. The antibody or binding fragment thereof according to claim 1, wherein the antibody is human-derived.
3. The antibody or binding fragment thereof according to claim 1, wherein the antibody is a monoclonal antibody and/or a recombinant antibody.
4. The antibody or binding fragment thereof according to claim 1, wherein the antibody is an IgG or IgA antibody.
5. The antibody or binding fragment thereof according to claim 2, wherein the human is selected from the group consisting of a human suffering from a peanut allergy, a peanut-sensitized human without a clinically-relevant allergy, a human suffering from a peanut allergy that underwent immunotherapy, a human that has outgrown a peanut allergy and a human of unknown clinical history for peanut allergy.
6. The antibody or binding fragment thereof according to claim 1, wherein the antibody is capable of reducing, inhibiting or neutralizing allergen-mediated biological activity.
7. The antibody or binding fragment thereof according to claim 1, wherein the antibody is capable of reducing or inhibiting the binding of an IgE antibody to the food allergen.
8. A composition comprising at least two antibodies or binding fragments thereof, wherein at least one of the antibodies or binding fragments thereof is selected from the antibodies in claim 1.
9. A pharmaceutical composition comprising the antibody or binding fragment of claim 1.

10. The pharmaceutical composition according to claim 9, further comprising a pharmaceutically acceptable carrier.

11. A kit comprising (i) the pharmaceutical composition according to claim 9, (ii) at least one additional agent useful for treating peanut allergy selected from the group consisting of β-adrenergic agonist, antihistamine, corticosteroid, anti-IgE antibody, anti-IgE antibody binding fragment, peptide vaccine and a further antibody capable of binding to a peanut allergen.

12. A kit according to claim 11, wherein the β-adrenergic agonist is epinephrine.

13. A method of treating a peanut allergy comprising administering an antibody or binding fragment thereof according to claim 1.